THÈSE DE DOCTORAT

DE SORBONNE UNIVERSITÉ

Ice nucleating particles from natural sources and the modelling of their interactions with Arctic clouds

Noyaux glaçogènes d'origine naturelle et modélisation de leurs interactions avec les nuages arctiques

Thèse soutenue par

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Abstract

Ice nucleating particles (INPs) are aerosol particles capable of promoting the formation of ice crystals in clouds, etc.

Keywords : Aerosol-cloud interactions – ice nucleating particles – cold clouds – Arctic – polar regions – microphysics – modelling – climate

Résumé

Les noyaux glaçogènes (INP, pour *Ice Nucleating Particles*) sont des particules d'aérosols capables de favoriser la formation de cristaux de glace dans les nuages, etc.

Mots clés : Interactions aerosol-nuages – noyaux glaçogènes – nuages froids – Arctique – régions polaires – microphysique – modélisation – climat

Foreword

This thesis accounts for what has occupied me over the past three years, bla bla blahh

Acknowledgements & Remerciements

The end of a thesis marks, for the PhD candidate, the end of an era... blablablah

Table of contents

Acronyms and Abbreviations

AA Arctic Amplification

AO Arctic Oscillation

AOD Aerosol Optical Depth

AR Assessment Report

ARTofMELT Atmospheric Rivers and the Onset of Arctic Melt

BC Black Carbon

CAM Community Atmosphere Model

CCN Cloud Condensation Nuclei

CESM Community Earth System Model

CMAQ Community Multiscale Air Quality model

CMIP Coupled Model Intercomparison Project

CO Carbon Monoxide

CO_2 carbon dioxide

CRE Cloud Radiative Effect

CRM Cloud Resolving Model

ECS equilibrium climate sensitivity

ESM Earth system model

fPBAP fluorescent PBAP

GCB Global Carbon Budget

GCM General Circulation Model

GHG Greenhouse Gas

INAS Ice Nucleation Active Sites

INP Ice Nucleating Particle

IPCC Intergovernmental Panel on Climate Change

IPSL Institut Pierre-Simon Laplace

IR Infrared Radiation

ISWP Ice and Snow Water Path

IWC Ice Water Content

IWP Ice Water Path

LATMOS Laboratoire ATMosphères et Observations Spatiales

LW Long Wave

LWC Liquid Water Content

LWP Liquid Water Path

MOCCHA Microbiology-Ocean-Cloud-Coupling

MOSAIC Multidisciplinary drifting Observatory for the Study of Arctic Climate

MPC Mixed-Phase Cloud

MPOA Marine Primary Organic Aerosol

NAO North Atlantic Oscillation

NOx Nitrogen Oxides

O_3 Ozone

OC Organic Carbon

OM Organic matter

PBAP Primary Biological Aerosol Particle

PBL Planetary Boundary Layer

PIP Primary Ice Production

RCP Representative Concentration Pathway

R_{eff} Effective Radius

RH Relative Humidity

RHi Relative Humidity with respect to Ice

S_i Saturation with respect to ice water

SIP Secondary Ice Production

S_l Saturation with respect to liquid water

SO_2 Sulfur Dioxide

SOA Secondary Organic Aerosol

SPM Summary for Policymakers

SSA Sea Spray Aerosol

SSP Shared Socio-economic Pathway

SST Sea Surface Temperature

SW Short Wave

TS Technical Summary

WBF Wegener–Bergeron–Findeisen process

WG Working Group

WRF Weather Research and Forecast model

WRF-Chem Weather Research and Forecast model including Chemistry

WS Wind Speed

Contributions to the scientific community

This side chapter is intended to list, and shortly describe, the various contributions of my thesis and side thesis activities to the scientific community. Here are presented the corpus of publications for which I contributed.

Peer-reviewed publications

Thesis research outputs

- Published in *Atmospheric Chemistry and Physics*: **Da Silva, A.**, Marelle, L., Raut, J.-C., Gramlich, Y., Siegel, K., Haslett, S. L., Mohr, C., and Thomas, J. L. (2025): How to trace the origins of short-lived atmospheric species: an Arctic example, Atmospheric Chemistry and Physics. DOI: https://doi.org/10.5194/acp-25-5331-2025
- blablablah

Scientific collaborations

- Published in *Atmospheric Chemistry and Physics*: Lapere, R., Thomas, J. L., Favier, V., Angot, H., Asplund, J., Ekman, A. M. L., Marelle, L, Raut, J-C, **Da Silva, A**, Wille, J. D., Zieger, P. (2024). Polar aerosol atmospheric rivers: Detection, characteristics, and potential applications. Journal of Geophysical Research: Atmospheres, 129, e2023JD039606. https://doi.org/10.1029/2023JD039606.
- blablablah

Conferences, colloquia, and workshops

2023

- EGU general assembly 2023, Vienna, oral presentation:

Da Silva, A., Marelle, L., and Raut, J.-C.: Seeking the origins of Arctic ice nucleating particles with FLEXPART-WRF, EGU General Assembly 2023, Vienna, Austria, 24–28 Apr 2023, EGU23-6392, https://doi.org/10.5194/egusphere-egu23-6392,2023.

- blablablah

International and national science projects

- Climate Relevant interactions and feedbacks: the key role of sea ice and Snow in the polar and global climate system (CRiceS).

  The CRiceS project focuses on improving model predictions of the role of polar processes in the climate system that consists of the oceans, ice and snow cover, and the atmosphere. It is crucial to understand the role of the polar processes, such as feedback loops, in polar and global climate. One of the main ways scientists can improve our understanding of environmental change is to combine knowledge from different disciplines in a coordinated way.

  My work contributed to the work package 2 (Improved ocean-ice/snow-atmosphere processes in models) and to the core theme 2 (Aerosols and Clouds).

- blablabla

Teaching activities

Sorbonne Université

I took part in six different courses, teaching students from first year to master level, within two departments of the university (*Earth's sciences* and *Physics*). Below stands the detailed list of those courses.

Undergraduate

- *Terre, Climat, Environnement*: tutorial leading and elaboration of presentations topics for the new maquette of the course.

- *Le changement climatique et ses enjeux*: supervising of students talks about climate-society relations.

- *Matlab modelling project*: supervising of modelling projects for five groups of students in geology.
- *Physique des millieux continus* : charge des TDs et des résolutions de problème (RP).

Graduate

- *Statistics* : tutorial leading and project supervising in advanced statistics with R.
- *Rayonnement et télédetection* : supervising of the lidar practical work session.

Ecole nationale supérieure des études avancées (ENSTA) Paris

I have been tutorial leader for the transdisciplinary course named *Analyse systémique de l'Anthropocène* (Systemic analysis of the Anthropocene), for ingineering school students.

European winter school ERCA 2025

I have been project group leader for six international PhD students. I supervised the group that focused on aerosol-cloud interactions. The objective I set up was to highlight the effect of ice nucleating particles on Arctic clouds thanks to principal component analysis of in situ measurements, and to empirical orthogonal functions computed on reanalysis data (ERA5 and CAMS).

Climactions IPSL

Under the name of *Climactions*, gather researchers of the Institut Pierre-Simon Laplace (IPSL) laboratories in a dynamic of thinking how to reduce the environmental footprint of their research. Since its launch in 2019, Climactions has expanded its objectives and shifted its focus: it is no longer only concerned with the environmental impact of research, but also with how to make research practices resilient in a society increasingly affected by the consequences of 2 °C global warming. I have participated to this dynamic through different activities presented below.

Working group research core activities

During the first two years of my PhD, I hosted the IPSL working group about the evolution of the observational activities of the climate sciences community. The group gathered researchers of various fields among the IPSL community. Our monthly meeting aimed to communicate and think about the levers to make scientific observation activities more resilient and sufficient: from the conception of the next generation of the French fleet of oceanographic ships, to the future of space observation, by way of Antarctic polar bases. The outline of our work can be found at Climactions' working group on observations activities[1].

[1] https://climactions.ipsl.fr/groupes-de-travail/gt2-observations/

Villarceaux Agreement and tool box

I actively participated to the elaboration of a panel of almost 50 measures (named the tool box), intended to help the IPSL laboratories reducing their environmental footprint. They also aimed to help the laboratories getting more resilient against budget cuts and various kinds of physical restrictions that public research may face in the near future.

In parallel, we set up the *Villarceaux Agreement* which— in the manner of the Paris Agreement—states the recognition of the need for our institute to change, reduce and adapt its activities, as well as the commitment to do so in order to preserve good research conditions for the present and next generations of scientists. Today, the Agreement has been signed by seven laboratories of the IPSL. The text can be found here: Villarceaux agreement text[2].

Climactions at LATMOS

I have participated to the local LATMOS Climactions group. With Lola Falletti (head of the LATMOS Climactions group), we organised a special general assembly to present four measures that have been then voted by all the laboratories members (LATMOS Climactions web page[3]). The next year, I organised a webinar to prepare the IPSL Transform'Actions week, when all IPSL laboratories gathered to discuss and choose measures from the tool box that best suit their activities. During that week, I led the LATMOS discussions. All along the three years of my PhD, I have made the link between the LATMOS directors and the IPSL head about Climactions affairs.

Science communication and awarness raising

Ma thèse en 180 secondes 2023 contest

I participated to the francophone contest *Ma thèse en 180 secondes*, where PhD candidates from every fields have 180 seconds to present their thesis to a non-specialist audience. I did the Sorbonne Université final. My appearance has been subtitled `https://www.youtube.com/watch?v=pGzU3e5XTz0`here[4].

High school teaching

- Interventions about global warming and climate sciences in high school classes (*2nde*, *1ère* and *terminales*).

- Atmospheric physics workshop with *terminales* students specialized in mathematics.

[2] `https://cloud.ipsl.fr/index.php/s/e7YaN7JEqw5r9nK?openfile=true`
[3] `https://climactions.ipsl.fr/groupes-de-travail/latmos-climactions/`
[4] `https://www.youtube.com/watch?v=pGzU3e5XTz0`

RADICAL

On the impulsion of François Ravetta, former director of LATMOS, his PhD student Thomas Lesigne and have held a scientific mediation mission about the RADiomètre Infrarouge pour la Course Au Large (RADICAL), a small infrared radiometer that the LATMOS installed on a racing sail boat. The project is resumed in the Radical booklet[5] and kakemono poster[6] we produced for the occasion.

[5] https://drive.google.com/file/d/1CV0wxgG6JPSrtKX6_IJspliK54sUBc2E/view?usp=sharing
[6] https://drive.google.com/file/d/17i1LEQ7NheX6M8gCM3Y0SALK7NreZvb9/view?usp=sharing

Résumé long en Français

Contexte scientifique

Au sein du climat terrestre, les régions polaires sont connues pour être à la fois extrêmes et particulièrement sensibles, bla bla blahhh

...

Résumé des Résultats | Modélisation des noyaux glaçogènes arctiques

Question initiale	Réponse apportée par la thèse
Quels outils sont nécessaires à l'identification des sources d'émission des particules atmosphériques ?	La modélisation de trajectoires lagrangienne, associée à des séries temporelles de mesures (in situ, aéroportées ou satellitaires) et analysée à l'aide de la méthode du ratio améliorée, permet l'identification fiable des sources d'émission de noyaux glaçogènes (INP), applicable également à d'autres types de composés atmosphériques (**??**).
Quelles sont les sources d'émission des INP arctiques ?	Les INP observés dans l'Arctique central proviennent probablement de l'intérieur même de la région arctique, avec un possible transport depuis l'Asie continentale. Deux principales régions sources ont été identifiées : les côtes russes des mers de Barents et de Laptev, ainsi que les côtes du Groenland. Les INP actifs à haute température (i.e. $-15\,°C$ et au dessus), présentent des origines océaniques (**??**).
Quelles sont les espèces d'aérosols importantes pour la nucléation de glace dans les nuages arctiques ?	Les principales sources d'INP suggèrent une prédominance de particules minérales dans l'Arctique, mais l'étude des INP de la campagne MOSAiC actifs à plus haute température ($-10\,°C$ et $-15\,°C$) a mis en évidence de possibles sources océaniques, indiquant la présence de particules organiques marines parmi les INP dits « chauds » (**??**).
Comment les différentes représentations de la nucléation de glace se comparent-elles dans un modèle régional ?	Les particules organiques simulées, combinées à des paramétrisations adaptées de nucléation de glace, produisent des concentrations d'INP plausibles. Les concentrations mesurées de PBAP fluorescents (fPBAP), converties en INP à l'aide de paramétrisations de nucléation, donnent toutefois des niveaux trop faibles pour soutenir l'hypothèse d'INP dominés par les PBAP (**??**).
Est-il possible d'utiliser une paramétrisation générale pour les INP arctiques ?	Oui, mais cela conduit probablement à des résultats biaisés, en particulier avec les paramétrisations dépendantes de la température (**??**).

Résumé des Résultats | *Suite*

Question initiale	**Réponse apportée par la thèse**
Est-il possible de représenter fidèlement les INP arctiques avec les connaissances actuelles, et comment procéder ?	L'émission de traceurs d'INP constitue une approche efficace pour reproduire des niveaux cohérents de concentrations d'INP. Les concentrations de l'ordre de $10^{-1}\,L^{-1}$ sont rares dans l'Arctique central pour des INP actifs à des températures supérieures à $-20\,°C$ (**??**). Une fois intégrée dans la microphysique, cette approche prédit des concentrations plus faibles que les paramétrisations générales, notamment dans les régions les plus isolées de l'Arctique (**??**).
La présence d'INP dans l'Arctique est-elle importante pour la formation et les propriétés des nuages ?	Les nuages simulés se révèlent très sensibles à la manière dont les INP sont représentés. Le traitement des INP dans les schémas microphysiques peut induire des impacts majeurs sur la fraction nuageuse et sur la phase des nuages (**??**).
Les INP ont-ils un effet significatif sur le bilan radiatif arctique via leurs interactions avec les nuages ?	Les travaux menés n'ont pas permis d'apporter une réponse définitive à cette question. Cependant, l'influence suggérée des INP sur les nuages implique une modification importante du bilan radiatif nuageux à la surface (**??**).

I Introduction to Arctic aerosol and cloud physics

"From this distant vantage point, the earth might not seem of any particular interest. But for us, it's different. Consider again that dot. That's here. That's home. That's us. On it everyone you love, everyone you know, everyone you ever heard of, every human being who ever was, lived out their lives. The aggregate of our joy and suffering, thousands of confident religions, ideologies, and economic doctrines, every hunter and forager, every hero and coward, every creator and destroyer of civilization, every king and peasant, every young couple in love, every mother and father, hopeful child, inventor and explorer, every teacher of morals, every corrupt politician, every "superstar", every "supreme leader", every saint and sinner in the history of our species lived there – on a mote of dust suspended in a sunbeam."

Carl Sagan

Chapter content

I.1 Introduction to Earth's climate, or what does the earth get from the Sun

This is the early morning of month September 1572. A slight breeze blows through the birches' foliage at Herrevad Abbey, yet untouched by the sea sprays of the Öresund strait, just fifty kilometres westward. Merging with the whisper of aspen leaves, the scraping sound made by the moving discs of Sophia Brahe's astrolabe alternates with the rustle of the pages of her observation notebook. The latter, full of azimuth and declination coordinates immersed within orbit sketches, was a pearl among the Renaissance astronomers. Her elder brother had left a few hours ago for his bed, but as for her, she could not leave the perfect conditions that this night offered. Her care and rigour, that she developed in astronomical observation and kept all her life, is what led the Brahe sister and brother to the discovery, a few month later, of the *Tycho's supernovae*. Their observation of this "new star" has been the cornerstone of the collapse of Aristotle's dogma of the immutability of the supralunar world. But that night, Sophia's mind was wondering far from antiquity's theories about the Universe. She was trying to make the most of this new moon and cloudless late summer night, with in her mind the completion of the tracking of two years of Mars' apparent retrograde motion. But when the sunlight started reaching the edge of horizon, Sophia got bothered in her last series of measurements. Even though she followed all night long the rise and cruising of the red planet, she just lost its track as it started its setting. First, she was tempted to blame her sleepy eyes, but she knew that the fresh morning breeze of Scania–present-day southern Sweden–always successfully kept her well awake. No, something less usual was happening, and it had nothing to do with the young astronomer's eyes. As she focused her gaze on the area where Mars should have been, she got intrigued by what she saw. No planet nor star was visible there, and it could have seem normal since the Sun was rising, but Venus and the Cancer constellation were still dominating the sky, although they were lower on the horizon. The image in front of her was very faint, but she could not be mistaken: the dark blue cloth of starry sky was masked by the perfectly round disk, yet very pale, of the full Moon. How could it have appeared while the whole night was clear of any moon? Sophia checked her notebook, looking for the lunar calendar she was keeping. No mistake, the Moon had just finished her twenty height days cycle. It had to stay invisible for a whole day before growing again as a crescent, waiting for the Sun-Moon-Earth triptych to get out of syzygy. But there was no sign of any slight slice of young Moon. Instead, the round and well known face of the earth's satellite was unveiling itself, as a nocturnal bird of prey casually casting its shadow after having caught its reckless meal within the forest humus. Without understanding it fully, Sophia was looking at her own reflection.

It is hard to know if Sophia Brahe could actually be surprised by the appearance of a full Moon disk on a new Moon day. Despite the fact she is recognized as the great assistant of her famous brother[1], the knowledge about her appreciation of physical causes is barely documented. However, her young age at the time–her birth being known to take place between 1556 and 1559––could let us imagine a teenager observing a new Moon for the first time, especially since this phenomenon is not as trivial as one could think. Getting back to the basics: the Moon appears bright to us because its surface reflects sunlight toward the earth. The apparent shape of this reflection evolves as the Moon rolls around the earth, affecting the overall configuration of this three-body system. When the Sun, the earth and the Moon are aligned, with the planet standing in between, the illuminated side of the Moon faces the earthlings who see it as a circular disk. Half a lunar cycle away to that full Moon configuration, stands the new Moon's. In fourteen days, the satellite has made its way to the other side of the earth, which makes it the middle piece of the trio. The Sun now lights the back of the Moon, and no sunlight is reflected towards the earth. Consequently, the new Moon nights are reputed to be perfect for astronomical observations, since only the observed objects are sources of light. The invisible Moon bothers no astronomer, unless it does... Here lies the enigma that caused Sophia's astonishment when she saw the complete Moon's visible face during that Moonless night. She could not see Mars any more because it got caught behind the invisible Moon (which is itself, a pretty rare event). But when the Sun starts rising and therefore, lights the earth surface, the latter partially reflects the solar rays towards space. Some of this reflected sunlight eventually reaches the Moon, revealing it to any attentive observer, like Sophia. This ability to reflect light is called the albedo. Measured as the fraction of reflected over incident short wave radiation (mostly known as solar radiation), the earth albedo reaches the temporally and spatially averaged value of 0.3. In other words, $30\,\%$ of the energy coming from the Sun and reaching the earth is reflected by the latter's atmosphere and ground. Which leaves $70\,\%$ of it to be absorbed by the planet.

This Sun's energy is how the earth gets its power. It is the energy that fuels the great majority of the processes that happen on Earth. From the plants in their photosynthesis efforts, to the solar panels that mimics the vegetal genius, including the uplifting of air parcels that makes the wind blow, which, when combined with the evaporation of sea water, makes thunder storms and hurricanes. This solar energy is not only the source of so-called renewable energies, but is at the root of the fossil fuels that have made the modern civilisation.

The first law of thermodynamics states that, in a closed system, the energy is conserved. Thankfully, the Earth is not a closed system, if only because of the energy it gets from the Sun. But in physics, an open system that gets energy without emitting any is closer to a boiler than to a vivarium. Therefore, since it is pretty uncommon to observe spontaneous boiling in the earth's oceans, one can assume that the planet is somehow getting rid of its energy. When a third of the incoming energy from the sun is reflected thanks to the planetary albedo, the rest

[1]Tycho Brahe was a 16th century Danish astronomer and alchemist. He is known for the observation of the supernova SN1572, and for his extremely precise sky observations from his observatory of Uraniborg, although they were performed with the bare eye.

is re-emitted as radiative emissions. Following Planck's law of thermal emission, any body of non zero temperature emits electromagnetic radiation; the emission wavelength depending on the body's temperature. The earth's surface temperature under the reign of humans averages around $13.5\,^{\circ}\mathrm{C}$[2], which corresponds to the brightness temperature in the middle of the infrared emission spectrum. Thus, the energy exits the system as infrared radiative emissions (IR). Because both inward and outward energy exchanges happen respectively as short wave (SW) and long wave (LW) radiation, we often speak of the Earth's *radiative budget*. In order to stay at an overall constant temperature, the Earth's system has to emit *exactly* the amount of energy received from the solar star. If the incoming energy amount (SW) is lower than the outgoing energy flux (LW), the planet gets cooler. Conversely, the slightest positive imbalance in the budget drives slowly but surely the system to warm (see Sect. I.2.3 for an example in real conditions), until the long wave radiations grow enough to match the incoming short wave radiations (Fig. I.1).

From this perspective, an apparent paradox–or rather a fundamental question–arises. If the energy received from the Sun is entirely re-emitted into space, it is hard to consider solar energy as a fuel–since it is not consumed–nor the Sun as a source. Consequently, the question is: what does the earth actually get from its star?

We said herein above that the solar energy was the source of every movement, except geological, on Earth. But yet, the energy neither dissipates nor accumulates in the system: it stays approximately constant by evacuating the amount simultaneously received. In order to lift the paradox, it is necessary to summon the second law of thermodynamics. First introduced by Sadi Carnot (1796-1832), and rectified by Rudolf Clausius (1822-1888), the Carnot's principle states that in any transformation, the energy implied changes its intrinsic qualities, losing some of its ability to transform. This lost, has been named *entropy* by Clausius, and writes S. Later, Josiah Willard Gibbs (1839-1903) introduced the concept of free energy, or available energy. Noted G–or F when relating to the Hermann von Helmoltz (1821-1894) potential–, the free energy points out the capability of physical transformation, and thus is related to low entropy states. Quantitatively, it is the maximum amount of work a thermodynamic system can produce. In the physics of phase transitions, G is the key quantity. In its modern form, the Carnot's principle, or second law of thermodynamics, states that, during any transformation within a closed system, the entropy can not decrease, and most likely increases. This implies the vanishing of free energy, replaced by bound energy. While the first law teaches the conservation of energy, the second states its systematic *degradation*. Unlike the Newtonian description of motion, the thermodynamics introduces an immutable irreversibility of physical transformations; it gives a one-way direction to Arthur Eddington's arrow of time. Additionally, it echoes–without contradicting it–the Lavoisier's principle, first pronounced at the dawn of the first French republic. Its famous statement is resumed as follows: "Rien ne se perd, rien ne se crée, tout se transforme."[3]. Even

[2]The year 2024 was the first to reach $1.5\,^{\circ}\mathrm{C}$ above the $13.7\,^{\circ}\mathrm{C}$ of the 1850-1900 global average.

[3]"Nothing is lost, nothing is created, everything is transformed."

though it was deduced from chemical experiments, and despite the fact it remarkably suits the late 18th century French politics, in order to fully apply to energy it needs to be amended with the notion of degradation: “[...] se transforme et se dégrade.”. One should hope that modern politics is not ruled by this updated version.

Thus, the Sun provides energy to the earth of course, but the latter does not keep it. What the planet does actually get and consume is the low entropy provided by the star. Beyond being a formidable nuclear power plant, the Sun is our one and only source of free energy, helping the earth keep its local entropy equilibrium. But it is only through life development that this free energy can turn into low entropy.

At his time, the physicist Erwin Schrödinger (1887-1961) was already throwing a critical eye on the way human societies were behaving toward Nature. Being recognized as a great physicist by his peers, he allowed himself to extend to biology his knowledge in physics. By looking at life on Earth, he understood that it was the embodiment of negative entropy, or *negentropy*, in the Universe: life feeds on solar free energy to develop and grow in organisation (i.e. as *organisms*), and doing so consume energy with local negative entropy balance.

The phenomena and processes presented, investigated and studied in this thesis all work with the free energy that comes from the Sun. So do the tools and techniques deployed to perform the three years of research presented in the next chapters; they all consume free energy, and doing so increase the earth's system entropy. This artificial increase in Earth's entropy contributes to destabilise the natural balance of things that allowed the living conditions we know to set in. In the next section, we will see how the dramatic increase of entropy by human activities threaten life conditions on Earth, whilst the research we do is often motivated by an improved knowledge of that phenomenon of destruction.

As living beings, should we not strive to behave like ones, i.e. by keeping a low entropy balance? and as advanced thinking animals should we always wonder: *“I am accelerating or slowing down the increase of entropy?”*

I.2 The Anthropogenic disturbance

> *“Human beings, who are almost unique in having the ability to learn from the experience of others, are also remarquable for their apparent disinclination to do so.”*
>
> – Douglas Adams

In 1980, the astronomer and famous science populariser Carl Sagan, was introducing the Cosmic Calendar for the first time. The idea was to squeeze the entire history of the Universe––from the Big Bang to nowadays–within one Earth year. In this calendar, the Big Bang happens at the first second of January 1st, while my PhD viva voce takes place at exactly midnight on December 31st[4], one year later. On that time scale, the Milky Way forms on May 6th, the Sun on August 31st. Fourteen days later Earth forms, and only eight days later the first life forms appear. But one has to wait until December 1st to see oxygen in the atmosphere. The reign on dinosaurs ends on the penultimate day of the year, and humans get differentiated from apes at 8 pm on the 31st. Homo Sapiens appears at 23:56, and finally Sophia Brahe was born at the beginning of the last second of 23:59. The last four hundred years of human civilisation fits in the very last second of the Cosmic Calendar. The most astonishing scientific and technological progresses along with the deadliest wars happened during that last second. During half of that microscopic fraction of the Universe's history, the human population grew of one order of magnitude. But most importantly, that last fraction of second sheltered an event that has never happened in the whole history of the earth, or of any other planet as far as our knowledge goes. Over the last two hundred years, a single species reached the power, then overpassed it, of astronomical and geological forcers. The 21st century started with human societies capable of altering the climate and jeopardising greatly the life conditions on Earth.

In this section, we will go through the 2025 global warming state of play and the scientific evidences that support it. Then we will review the main causes and mechanisms of the phenomenon and its expected evolution. Finally, we will discuss the implications of such an unprecedented experiment for our research practices and topics.

I.2.1 An Earth's climate outline

The climate is commonly defined as the meteorological conditions, averaged over a long period that can extend from months to millions of years, depending on intention. But defining the climate as a statistical object is reductive. The term climatologist does not only refer to weather statisticians. Climatologists include researchers specialists of ocean currents, clouds, volcanoes or forests. A climate is what something is acclimatised to; on Earth, this something are living beings. Consequently, climatology needs to be taken as a whole. It is not possible to understand Earth's climate without studying the entirety of what it constitutes: the living condition on Earth.

The climate can approached as a system made of five components that interact continuously with each others: the atmosphere, or the layer of gas and in suspension particles that surrounds the planet; the hydrosphere, or the water that compose oceans, rivers, aquifers and clouds; the cryosphere, or every frozen surface including sea ice, glaciers, snow cap and permafrost; the lithosphere, or the rigid envelop of the planet; and the biosphere or the ensemble of living beings, from bacteria to humans, including all plants, animals and microorganisms.

This thesis will primarily focus on the atmosphere. Nevertheless, its interactions with the four others components of climate will be time to time at the heart of the subject. The cryosphere will

[4] That is not the actual date and time of the defence. Beware of mixing the Cosmic Calendar with the Gregorian one.

be the common thread, as the Arctic is the permanent landscape where the action takes place. Since the tropospheric clouds are major actors in the following chapters, the hydrosphere will naturally accompany the general intention. As for the lithosphere and biosphere, they will take action about aerosol particles compositions and sources, and will not exit the scene until the end. In that way, this PhD work can be seen as a study of the interrelations of those five climatic actors, with a focus on the causal links they bear.

Beyond the interactions within the climate system, the earth interacts with its outer environnement, namely: outer space. As mentioned in the previous section, the earth receives free energy from the Sun, and has to emit toward space the same amount of degraded (i.e. high entropy) energy. This happens through what is called the radiative budget of the planet. One third of the incoming solar radiations is reflected by the ground and the atmosphere (due to the albedo); the remaining is absorbed by the system. In response, the planet (through its five climate components) emits infrared radiations toward space. Except a small part–that goes through what is called the atmospheric window–, all this long wave radiation is absorbed by the atmosphere. This absorption is due to greenhouse gas (GHG), of which, on Earth, the most important is water vapour. GHG are gas that have absorption rays in the telluric infrared spectrum, or in other words, they can absorb the long wave radiations emitted at Earth's temperature. The atmosphere, warmed up by long wave absorption, then re-emits this energy in all directions, toward ground and outer space[5].

An increase of GHG concentrations leads to an upward shift of the radiatively opaque portion of the atmosphere, and thus of the altitude at which the atmosphere radiates toward space. Consequently, the LW emission temperature of the earth gets cooler, causing less energy to escape the system. It creates an imbalance in the radiative budget of the whole system, which leads to its warming. This radiative imbalance is also called a radiative forcing. This forcing drags the climate system toward a new energy equilibrium–with a warmer average temperature–allowing stronger long wave radiation capable of overcoming the additional infrared absorption by GHG. At the first order, as long as the incoming solar radiation are relatively stable (what they are), the concentrations of GHG drive the average temperature on Earth. Consequently, if the GHG concentrations keep on increasing, so would do the temperature of the new equilibrium.

I.2.2 Evidence of Global Warming

> *"This summer is the hottest of my life!–No son, it is the coolest of the rest of your life."*
>
> – Anonymous climate scientist on the beach

The Intergovernmental Panel on Climate Change (IPCC), is a scientific body established to provide comprehensive assessments of the scientific, technical, and socio-economic information

[5]It is interesting to notice that the actual surface power density of long wave emissions from the atmosphere toward ground is about $324\,\mathrm{W} \cdot \mathrm{m}^{-2}$, more or less the double of short wave incoming radiations at the surface.

relevant to understanding human-induced climate change and its impacts. By critically evaluating the existing literature and synthesising findings to inform climate action by various stakeholders, including governments and international organisations, the IPCC serves as a critical platform for synthesising climate science, fostering collaboration across disciplines, and providing evidence-based insights to guide global climate action. Its rigorous assessment process and commitment to transparency and inclusivity are fundamental to its role in addressing the challenges posed by climate change. The IPCC is organised into three main working groups, each focusing on different aspects of climate change:

- Working Group I (WG1): Concentrates on the physical science basis of climate change, assessing the scientific literature related to climate systems and their changes.
- Working Group II (WG2): Focuses on impacts, adaptation, and vulnerability, evaluating how climate change affects ecosystems and human systems.
- Working Group III (WG3): Deals with mitigation strategies, analysing options for reducing greenhouse gas emissions and enhancing carbon sinks.

In order to understand what is global warming and what are its causes and consequences, we will rely on the work of WG1. More specifically, we will use the outputs of the sixth Assessment Report (AR6), released in 2021 for WG1, and the final report in 2023. It is the most recent AR, the next one being expected for 2028-2030.

Figure I.2 presents a selection of graphics from different sources that illustrates the recent evolution of Earth's surface temperature. Panel I.2A shows the reconstructed and observed global temperature evolution for the last 2 000 years. It is displayed as an anomaly compared to the average temperature in 1850 to 1900. After a slow cooling that started around year 1000, the period 1850-2000 present a sharp warming that get much higher than the variability of the eighteen previous centuries. In fact, during the 20th century, the temperature reaches its higher level since at least 100,000 years.

Panel I.2B zooms in on the period 1850-2000. In black is represented the detail of the temperature series of panel A, but it is compared with two climate simulations. In green is the simulated global temperature in a climate with only natural forcing, namely solar and volcanic influences. Despite multi annual variations, the temperature remains stable around its level of 1850. In light brown, the temperature curve is from a climate simulation that takes into account the anthropogenic factors (c.f. Sect. I.2.3). Although it does not match perfectly with the observation curve, this second simulation reproduces much more accurately the actual evolution of temperature. At the bottom of the graph are represented the *climate stripes*, that depict in a different way the temperature evolution during that period.

Panel I.2C is the ranking of the ten warmest years since the beginning of measurements, until 2023. The latter stands at the first place, by almost 0.2°C above 2016. What is remarkable with this ranking, is that the ten warmest years are also the last ten. In other words, even though every year does not beat the record, it is most likely that it will be overtaken soon enough. 2024 broke the record of 2023, being the first year to beat the $1.5\,°C$ warming level.

Finally, panel I.2D is the seasonal cycle of global temperature for the three years of my PhD: 2023, 2024, and 2025 until I write these lines. Along with those three years, is highlighted the year 1992 when the third Earth summit happened. At that occasion, the United Nations Framework Convention on Climate Change (UNFCCC) was established, along with the Convention on Biological Diversity (CBD). These enter into the commitment that the United Nations took that year: to protect future generations by preserving the living conditions on Earth. The curves show that, since then, the global warming that took place between 1850 and 1992, more than doubled. But was it even stoppable?

I.2.3 Root causes or, of natural climate evolution timescales

"Crisis? What crisis?"

– Supertramp

I.2.3.1 Drivers of climate variability

The earth's climate is in essence an evolutive system. Its complexity, as glimpsed in the previous sections, makes it a chaotic system that reaches equilibrium from time to time. Among the near infinity of climate drivers, some of them deserve to be cited as the main causes of climate evolution. First of all, the orbital parameters of the earth have be theorised by Milutin Milanković, James Croll and Joseph-Alphonse Adhémar as the pacemaker of the climate (**hays_variations_1976**). The longest and most impactful orbital cycle find its source in the variation of the earth's orbit eccentricity, with a main period of $400\,000\,\mathrm{years}$, and a secondary period of $100\,000\,\mathrm{years}$. The second is the obliquity (or inclinaison of the rotation axis with respect to the ecliptic plane[6]) variation, which cycles every $41\,000\,\mathrm{years}$ and acts on the seasons strength. A low obliquity causes a weak seasonality, while a strong one strengthen the seasonal gradient. Finally, the shortest is due to the equinox precession, with a period of $26\,000\,\mathrm{years}$, and can cause inequality in seasons magnitudes among the two hemispheres. The combinaison of these has caused the great climate variability over the last hundred of thousand years. These climate variations always have a main parameter, namely the surface temperature. Because all of the five climate components are sensitive to thermodynamics, the evolution of temperature causes a cascade of changes creating feedback loops. Some of them can act as restoring force, preventing phenomena to grow strong, and are called *negative* feedback loops. Others can fuel the phenomenon that initiated them, like vicious circle, or what is called a *positive* feedback loops.

Among other major drivers, tectonic activity and volcanism can induce non-cyclic climate forcing. Large eruptions inject significant amounts of ash and sulphur dioxide into the strato-

[6]The plane of the earth's (and most planets') orbit around the Sun.

sphere, leading to temporary cooling effects due to increased albedo and reduced solar radiation reaching the surface. Important injection of water vapour into the stratosphere can also happen during volcanic eruptions. It is estimated that the 2022 eruption of Hunga Tunga underwater volcano blasted into the stratosphere 10% of the total stratospheric water vapour, with complex effects on the climate (**chen_impact_2025**). Eruptions effects can last from a few days to a few months for the most common events, but very large eruptions have induced major changes to the past climate, with important consequences on living species.

Eventually, the Sun forces periodic variations to the climate. Putting aside the short yearly and very short daily variations of solar radiation due to the orbital motion of the earth, the Sun presents an activity cycle of eleven years in average. The associated variations in energy flux reaching the earth's surface during a cycle is about 0.1 to $0.2\,\mathrm{W}\cdot\mathrm{m}^{-2}$. Meanwhile, the radiative imbalance induced by anthropogenic activities is estimated at $2.7\,\mathrm{W}\cdot\mathrm{m}^{-2}$ (**intergovernmental_panel_on_climate_change_climate_2023**). Longer solar cycles have been theorised and partially observed, and are candidates to explain past climate events. Nevertheless, the magnitude of such variations always stay around a tenth of a Watt per square meters, one order of magnitude lower than the current anthropogenic forcing.

I.2.3.2 The anthropogenic forcing

As mentioned in the preamble of this section, humanity has gotten, over the last hundred years, a power of action equivalent to the natural drivers presented herein above. Despite its extreme power of destruction on ecosystems (see Sect. I.2.5), its main impact comes from its ability to change the atmospheric chemical composition. Figure I.3 depicts the evolution, from the start of the Gregorian calendar until today, of the atmospheric concentration of carbon dioxide (CO_2). The curve shows how stable this concentration was over the past two thousand years. During the first three quarters of that period, civilisations have risen and collapsed, wars have roared, periods of famine and plenty have succeeded one another. Notwithstanding, the curve remained still until the late 18th. What happened then is well known and understood: the first two industrial revolutions happened.

The term industrial revolutions refers to the historical periods characterized by rapid industrial growth and profound socio-economic impacts. We call *first* industrial revolution (1760–1840), the dramatic increase of England's industrial production. Supported by the invention of the steam engine and the great English coal stocks, this period of great social changes brought England to the top of the world's production rankings. After a few decades, these techniques got exported, and the revolution extended to Europe and later to other nations, including the United States. This second industrial revolution marked the commencement of a substantial escalation in the use of fossil fuels.

The combustion of hydrocarbons (which fossil fuels are composed of), combines the dioxygen of air with the carbon of the long molecules of fuel. Among other chemical co-products of the combustion (including particulate matter), CO_2 is emitted. The absorption spectrum of CO_2 contains rays in the infrared wavelengths, right into the emission spectrum of the telluric emissions of the earth. Therefore, atmospheric CO_2 tends to absorb and block the outgoing long wave radiations. An increase of CO_2 atmospheric concentration induces an imbalance in the radiative budget, causing a warming until a new equilibrium is reached. However, while the concentration is increasing, the radiative budget stays imbalanced and the warming goes on. This additional greenhouse effect is the root cause of the observed global warming.

Figure I.4 gives an overview of the annual CO_2 emissions per region and activities (panel A), as well as the evolution of primary energy consumption by sources (panel B). Both graphs highlight the striking change of pace after WWII. This acceleration was prepared by the slow but steady development of fossil fuels within the industry. Panel B shows that the increasing use of coal since the middle of the 19th century, has been joined by the rapid development of oil in the first half of the 20th century. Simultaneously, but slower, the natural gas[7] rose in the global energy mix.

It is commonly mistaken that primary energies replace each other, as part of a continuous and systematic technological improvement. But as illustrated by Fig. I.4 B, the energies really place themselves on top of each others. **fressoz_pour_2021** demonstrates that they do not only pile up, but also stimulate the use of each others, in a symbiotic dynamic. Coal development did not suppress wood consumption, but rather stabilised it because of the need of coal mines and railways for wood props. Oil never replaced coal, but rather enhanced its production thanks to the development of terrestrial and marine transportation it allowed. Gas piled as an extra primary energy, but the inherent constraints of gas phase fluids technically prevent it to be widely used instead of oil or coal. Finally, the other energy vectors stay marginal, and have low to no contribution to the reduction efforts of global fossil fuel consumption. The new renewable energy vectors, namely solar and wind, even develop in a symbiotic relation with fossil fuels, as their production greatly relies on the oil, coal and gas industry. Nowadays, $80\,\%$ of the energy used in the world still comes from fossil fuel combustion.

Figure I.5 shows the estimated radiative forcing of the main anthropogenic atmospheric species, and the associated effect on global temperature. In this context, the radiative forcing is estimated as the change in the top-of-the atmosphere radiative power flux (in $\mathrm{W \cdot m^{-2}}$) caused by changes in atmospheric composition since the pre-industrial period. While CO_2 and methane (CH_4) are the main contributors to the total additional radiative forcing, some other species have a net negative radiative forcing. Sulfur dioxyde (SO_2) shows a negative forcing around $-1\,\mathrm{W \cdot m^{-2}}$. Its effect splits into two phenomena. The weakest but best quantified is due to the direct aerosol-radiation effect (c.f. Sect. I.4.1). The other is stronger but presents high uncertainties. It is caused by the interactions of sulphate aerosols with clouds, or what is referred as indirect aerosol effects.

Panel C of the figure illustrates the evolution in time of these forcings, highlighting the

[7] Not more natural than coal or oil, natural gas is a side product of the latter two, and therefore belongs to fossil fuels as well.

discrepancy among the concerned species. For instance, even though CH_4 is a strong GHG, it is chemically reactive in the atmosphere. Thus, its residence time in the atmosphere is relatively low. Conversely, CO_2 is chemically inert, and therefore very stable, which makes its lifetime very long. After 90 years, CO_2 has lost less than 20 % of its warming impact, while less than 5 % of CH_4 remain. The total negative forcing of CH_4 suffers the same evolution, with less than 5 % remaining after 90 years.

This demonstrates that the global scale experience that is the anthropogenic induced global warming does not allow any easy reversal. The tenth of billion of tons of CO_2 emitted each year by human activities are there to last. A sudden stop of GHG emissions would not permit a come back to the pre-industrial climate, but it would help avoiding things getting worst. Until radical changes are engaged in order to drastically reduce those emissions, the only certainty is that tomorrow's climate will be warmer, and Earth's living conditions increasingly degraded.

I.2.3.3 An unprecedented rate of change

What makes the current global warming so specific, is its rate of change. In Earth's history, climate warmings and coolings happened with magnitudes of one to ten degrees. Over the last 800 000 years, the Milanković cycles induced an alternance of long ice ages and short tempered periods like the Holocene we are in. At the last glacial maximum (LGM), 21 000 years ago, the average surface air temperature was 9°C, which is 4 to 5°C lower than the temperature of 1900. The warming that brought the global temperature up to 13°C took 5 000 to 10 000 years to happen. In comparison, the temperature rose of 0.75 °C between 1900 and 2000; since year 2000 this warming almost doubled, and is now near 1.5 °C. Those correspond to warming rates respectively seven and thirty times higher than the last deglaciation[8].

I.2.4 Expected future evolution of climate

Apart the work of understanding the past and current climate changes, the scientific community–from which the IPCC pull their evidence–also focuses on climate forcasting. In other words, the IPCC reports have also the charge of depicting the face of the future climate from the last advances in climate sciences. In their last report, the AR6, the IPCC introduced a new scriptwriting tool called Shared Socio-economic Pathways (SSP). SSPs are representative of the human societies responses to the global warming, and are complementary to the more classic Representative Concentration Pathways (RCP) of the past ARs, which only considered the GHG emissions and the associated radiative forcing. The SSP nomenclature works like this: in SSn-X, X refers to the radiative forcing reached in 2100, expressed in Watts per square meters, similarly as the RCP convention. The number n, accounts for the type of global collective response to the global warming, giving a narrative dimension to the scenario. Those are defined as follows:

[8] Assuming a lower bound rate of 5°C over 5 000 years.

- SSP1 – Sustainability: This pathway envisions a world focused on sustainable development, characterised by inclusive growth, reduced inequalities, and a strong emphasis on environmental protection. It assumes significant progress in technology and education, leading to low greenhouse gas emissions and high resilience to climate impacts.

- SSP2 – Middle of the Road: This scenario represents a continuation of current trends, where socio-economic development follows a moderate trajectory. It combines elements of sustainability and fossil-fuel-based growth, resulting in moderate emissions and adaptation challenges.

- SSP3 – Regional Rivalry: This pathway depicts a fragmented world marked by regional conflicts and nationalism. Economic growth is slow and cooperation on climate issues is limited, leading to high emissions and significant adaptation challenges due to a lack of resources and technology transfer.

- SSP4 – Inequality: This scenario highlights a world characterised by stark inequalities, where socio-economic disparities hinder effective climate action. Wealthier regions are better able to adapt, while poorer regions remain highly vulnerable, resulting in high emissions and considerable difficulties in ensuring climate resilience.

- SSP5 – Fossil-fueled Development: This pathway envisions rapid economic growth driven by fossil fuel consumption. It assumes the absence of stringent climate policies, resulting in very high greenhouse gas emissions and severe climate impacts, with limited adaptive capacity.

By just comparing the actual emissions of greenhouse gases to the depicted SSP scenarios, the road we collectively take is located around SSP3. Figure I.6 shows how difficult the way is to achieve the necessary emissions reduction.

Figure I.7 graphically summarises the projected increases in temperature from 2020 to 2100. The top-right panel shows projections from five SSP for three ranges of Equilibrium Climate Sensitivities (ECS). Climate sensitivity considers the temperature change associated with a doubling of CO_2 concentration in the atmosphere. At equilibrium, this sensitivity not only reflects the warming potential of greenhouse gases but also the feedback mechanisms within the climate system, including those related to atmospheric water vapour, clouds, and ice albedo. The estimates of ECS are based on general circulation models of the ocean and atmosphere and are still highly uncertain, which is why Figure I.7 includes a range of ECS to represent possible future changes.

The top-left panel shows the probability distribution of ECS levels. This distribution indicates that, for a doubling of atmospheric CO_2 relative to pre-industrial levels (the conventional expression of ECS), the expected warming ranges between 2.5°C and 4°C.

The projections corresponding to this ECS range are shown in the right panel "Mid-range ECS." It shows that the 1.5°C global warming limit, targeted by the Paris Agreement, will be exceeded during the 2030s under any SSP scenario. Depending on the scenario, the 2°C threshold (which the Paris Agreement aims to stay well below) will be crossed between 2040 and 2055 for the three least ambitious climate policy scenarios (SSP5, SSP3, SSP2) and will not be exceeded only under the two SSP1 scenarios. The two lower panels illustrate two types of consequences of global warming for three levels of warming: 1.5°C, 2.0°C, and 4.0°C, across different world regions. The first row relates to the Heat Warning Index. This index represents the number of days per year where temperature and humidity conditions reach dangerous levels for human health. Since the human body cools via sweating (with sweat evaporation helping to reduce body temperature), relative humidity plays a key role in tolerating high heat. For a given temperature, a lower humidity level makes the heat more bearable. The maximum tolerable temperature for a healthy young adult doing minimal physical activity is estimated at $50\,°\mathrm{C}$ with $10\,\%$ relative humidity or $35\,°\mathrm{C}$ with $70\,\%$ humidity. The maps show, for instance, that at $2\,°\mathrm{C}$ warming, regions within the intertropical zone would face a heat warning index of 30 to 180 days per year. Linking this to the projections in the top part of the figure, and assuming an SSP2-4.5 scenario, these effects are expected around the year 2050. In 2020, these regions housed more than one-third of the global population, raising concerns about future habitability in vast parts of the world.

The bottom row shows changes in extreme rainfall events. The values express the percentage increase in rainfall on the wettest day of the year. Climate warming leads to increased air humidity (according to the Clausius–Clapeyron relation, every $1\,°\mathrm{C}$ increase causes a $7\,\%$ increase in humidity), resulting in more intense rainfall in a warmer atmosphere. There are, of course, regional and seasonal variations to this trend, but it generally holds. Local variations can also be linked to changes in atmospheric circulation. Regional feedbacks are also possible; for instance, the presence or absence of vegetation (linked to rainfall changes) can, in turn, influence rainfall levels. By 2050, under the mid-range ECS, the least ambitious trajectories place us at warming levels between 2 and $4\,°\mathrm{C}$, while the most ambitious ones keep us between 1 and $2\,°\mathrm{C}$. It is clear that extreme heat and rainfall events remain much more contained below $2\,°\mathrm{C}$ than at $4\,°\mathrm{C}$, when entire regions of Africa or South America would face heat levels dangerous to human health for more than half the year.

I.2.5 The silent collapse

We did not mention it yet because it is not related to climate, but its importance is so immense that I cannot avoid dedicating a small section to it. Apart from the influence of human activities in disturbing the planetary climate, the first and most striking consequence of modern global capitalism is the organised destruction of ecosystems. The numbers are overwhelming:

- Over $75\,\%$ of flying insects populations have disappeared in protected areas over the last $27\,\mathrm{years}$ (**hallmann_more_2017**).

- Same analysis for the arthropods–the most diversified living group–, over three quarters have gone in a few decades (**seibold_arthropod_2019**).

- The global rate of extinction is estimated to be 100-1000 times greater than the background rate (**lamkin_challenge_2016**).

- The total disparition of living individuals, all species and groups combined, amounts to around $70\,\%$ (**ipbes_summary_2019**).

- The living planet index[9] has decreased of approximately $20\,\%$ in North America and Europe since the 70's, but far from our eyes it has already collapsed: $-55\,\%$ in South-east Asia and Oceania, $-64\,\%$ in Africa and $-94\,\%$ in South America (**ritchie_biodiversity_2022**).

The causes are known and understood. Almost all human development is achieved to the detriment of non-human living beings. In other words, human activities affect negatively $99.9\,\%$ of the total biomass on Earth (**bar-on_biomass_2018**). This appears clearly when looking at the distribution of mammal mass: $60\,\%$ are livestock, $34\,\%$ are humans, and $4\,\%$ only are wild mammals. As for birds: $70\,\%$ are chickens and other poultry, and only $30\,\%$ are among wildlife (**bar-on_biomass_2018**). Even though domestic animals now compose the greatest part of ground living vertebrates, their time on Earth is extremely short: just in France, $3\,\mathrm{million}$ animals are killed every day. Worldwide, this number goes up to $400\,\mathrm{millions}$ to reach 150 billion every year. This is without taking into account marine animals who represent $96\,\%$ of killed animals for human supply[10].

The Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) reminds us:

> *"While more food, energy and materials than ever before are now being supplied to people in most places, this is increasingly at the expense of nature's ability to provide such contributions in the future, and frequently undermines nature's many other contributions, which range from water quality regulation to sense of place. The biosphere, upon which humanity as a whole depends, is being altered to an unparalleled degree across all spatial scales. [...] Nature is essential for human existence and good quality of life. Most of nature's contributions to people are not fully replaceable, and some are irreplaceable."*

– ipbes_summary_2019

[9] The living planet index (LPI) is a measure of the state of the world's biological diversity based on population trends of vertebrate species from terrestrial, freshwater and marine habitats.

[10] See the list of references from the Food and Agriculture Organization of the United Nations (FAO) at `https://www.l214.com/animaux/statistiques-nombre-animaux-abattus-monde-viande//`.

We may see less insects flying, although Europe is spared compared to the rest of the world. We may hear less birds singing, although birds represent only a tiny fraction of total living species. We cannot hear it, nor see it, but the science is clear, that we are living a *silent collapse*, while the spectrum of the Sixth Mass Extinction hovers above the 21st century societies.

> *"Denying the crisis, simply accepting it and doing nothing, or even embracing it for the ostensible benefit of humanity, are not appropriate options and pave the way for the earth to continue on its sad trajectory towards a Sixth Mass Extinction."*

– cowie_sixth_2022

I.2.6 Implication for scientific research

By the time current PhD students are the age of their experienced supervisors, the climate would have evolved most likely as depicted on Fig. I.7. The geopolitical responses and evolutions of such a rapid change of living conditions can be expected to be of an unprecedented magnitude. What will be the implications for scientific research, for our social system, and most importantly: what should we do? These questions are addressed in the appendix, as a dedicated section (see **??**).

NB: From this point, the intention gets closer to the thesis research project by introducing clouds, their role in climate, the physics of their formation and evolution, as well as the physics of ice nucleating particles and the Arctic climate.

./chap1/images/radiative_budget.png

Figure I.1: Schematic of Earth's radiative budget. The yellow arrows indicate solar radiation; the red arrows illustrate long wave radiation. The values are expressed in $\mathrm{W \cdot m^{-2}}$; in green, the energy received by the earth, and in red, the energy emitted back to space. Note the role of the atmosphere and clouds, whose overall effect on the surface is of the same order of magnitude as the incoming solar radiation at the top of the atmosphere. Adapted from **kiehl_earths_1997**.

./chap1/images/warming_evidences.png

Figure I.2: Elements of proof for a human-caused global warming. Panel A shows the change in global surface temperature relative to the average of 1850-1900, as reconstructed (1-2000) and observation data (1850-2020). Panel B is a zoom of panel A with observed and simulated data, using human & natural and only natural factors (**masson-delmotte_summary_2021**). Panel C present the top ten of the warmer years in global average (source: https://www.climatecentral.org/). Panel D is the global air surface temperature yearly evolution of the three years of my PhD, compared to 1992, the year of the third Earth summit when the United Nations committed to protect the future generations (data from C3S/ECMWF, ERA5, Copernicus: https://pulse.climate.copernicus.eu/).

./chap1/images/co2_2k_ce.pdf

Figure I.3: The "Keeling Curve": this graph illustrates the evolution of atmospheric CO_2 concentration over the last two thousand years. Named after Charles D. Keeling who initiated the measurement project of continuous CO_2 measurement at Mauna Loa observatory (Hawaii, U.S.) in 1958 (**keeling_atmospheric_2017**), the data is completed by ice core measurements (**rubino_revised_2019**). The industrial revolution is clearly highlighted by the dramatic increase that started in the late 18th century. Data can be found here: https://keelingcurve.ucsd.edu/.

./chap1/images/world_energy_consumption.png

Figure I.4: Historical CO_2 emissions by regions and activities (A), put into perspective with the evolution of primary energy consumption by source. Data from *Our World in Data*: https://ourworldindata.org/.

./chap1/images/IPCC_AR6_WGI_TS_Figure_15_tot.png

Figure I.5: Effective radiative forcing (A) and associated additional warming (B) for different atmospheric species, from 1750 to 2019. (C) Change in global temperature, after 10 and 100 years, caused by a single year pulse of 2019 anthropogenic emissions. Graphics from **intergovernmental_panel_on_climate_change_climate_2023**.

./chap1/images/IPCC_AR6_SYR_Figure_2_5.png

Figure I.6: Trend from implemented policies (red line) compared to emissions scenarios that would allow to stay below 1.5°C (cyan line) or below 2.0°C (green line) of global warming. A scenario that takes into account the National Determined Contributions (NDCs) discussed during COP26 is also represented (dark blue line). The NDCs represent the free-of-result commitments made by countries to reduce GHG emissions and enhance resilience to climate change impacts. Graphics from **intergovernmental_panel_on_climate_change_climate_2023**.

./chap1/images/IPCC_AR6_WGI_TS_Figure_6.png

Figure I.7: Graphical abstract of the key aspects of the IPCC Assessment Report 6, Technical Summary. The upper panels are temperature projections for the five SSPs and three ranges of ECS. The lower panels are the responses of two climate indexes (heat warning index and change in extreme rainfall) for three levels of warming (1.5°C, 2.0°C and 4°C). The association of upper and lower panels allows to catch a glimpse of the timeline for future global warming consequences. Graphics from **intergovernmental_panel_on_climate_change_climate_2023**.

I.3 Clouds and microphysics

In the short film of Pier Paolo Pasolini *Che cosa sono le nuvole?*[11] (1968), two theatre puppets that have had their day, end up in a rubbish tip. Thrown on their back, they can just stare at the sky, and are found to be enthralled by what they see:

> What are those? asks one of the two puppets. Those are clouds, answers the other. – What are clouds? – Who knows? – How beautiful they are! How beautiful they are!

This question shelters many dimensions and can be explored in a multitude of manner, and by addressing it we gently get closer to the heart of the thesis. In this section, we will explore the physical nature of clouds through their meteorological definition, their formation and evolution dynamics, and by depicting the processes that intervene in the scientific work presented in the next chapters.

I.3.1 What are clouds

> *"Some say that black holes are the most complex objects in the Universe, although they're just regions of space-time from which the light cannot escape. On the other hand, a cloud is a region of the atmosphere where thermodynamic and dynamical conditions contribute to enabling the formation of liquid or solid, or both, water particles that manage to stay in suspension in the air during an adequately long time for an impressionist to paint them, until a stochastic cascade effect causes all this water to fall, unless it doesn't and stays in the sky preventing any astrophysicist from studying black holes."*

– A.D.S.

Earth's atmosphere contains at any moment, not less than 10^{25} hydrometeors, or water based particles in suspension. The majority are liquid water droplets, and the rest is ice crystals (**cantrell_production_2005**). In parallel, the overall cloud coverage on the planet tends to be of 60-70 %. In one hand, these numbers quantitatively describe the whole ensemble of clouds, but do not help to picture a single cloud. On the other hand, it is fairly easy to point out a cloud just by looking at the sky, although this approach is purely qualitative. Luke Howard (1772-1864), sometime considered as the *"Godfather of clouds"* for the stratus-cumulus-cirrus classification he set up (still used nowadays), took inspiration in the artwork of cloud representation, and more precisely among the romantic painters such as William Turner and Caspar David Friedrich (**howard_lxiv_1803**). Yet, their paintings were rarely scientifically accurate. An other romantic

[11] *What are clouds?*

painter, John Constable (1776-1837), used Howard's classification and scientific observations of his own to accurately represent clouds, before photography could immortalise the sky's tempers. Even though the bare eye approach is the most natural to the human observer, it most likely misses the majority of physical properties that make a cloud, a *cloud*.

The World Meteorological Organization (WMO) and the American Meteorological Society (AMS), do not agree on the cloud definition. The first defines it as:

> An aggregate of very small water droplets, ice crystals, or a mixture of both, with its base above the earth's surface, which is *perceivable* from the observation location.

The second gives a broader definition that includes non water clouds:

- A visible aggregate of minute water droplets and/or ice particles in the atmosphere above the earth's surface.
- Any collection of particulate matter in the atmosphere dense enough to be perceptible to the eye, as a dust cloud or smoke cloud (**world_meteorological_organisation_international**

The main distinction is the inclusion of non water particles aggregate in the definition of AMS, while WMO only consider water particles (droplets and ice crystals). It overlaps with the definition of aerosol particles (see Sect. I.4) which is rather uncomfortable. The second, and more subtile distinction, is about the cloud detection. AMS highlights the necessity of cloud to be visible to the eye. Conversely, WMO does not mention the naked eye detection, but rather speaks of *perceivable* aggregates. This includes instrumental observations, that are sometimes the only way to detect clouds. Indeed, the development of instrumental techniques for cloud observation has led to the discovery of perfectly invisible but stable aggregates of hydrometeors that follow cloud dynamics and radiative properties (**lesigne_extensive_2024**). Looking at clouds through instruments implies the use of sharp thresholds on physical variables. The set of parameters will then differ from a discipline to another. In radiation, the optical thickness will rule, while in cloud microphysics, the cloud particles characteristics (number and radius) and the liquid and solid water contents will be essential.

In **spankuch_what_2022**–where the cloud definition is extensively discussed–the authors try to define the modelled cloud like the following:

> From a modelling point of view a cloud could be tentatively defined as follows: a hydro-thermodynamic nonequilibrium system of the coexisting macrophases of water in a turbulent humid air atmosphere including (i) the metastable states of water (supersaturated vapour, undercooled water), (ii) the different hydrometeoric forms of water condensates, and (iii) primarily emitted and secondarily formed atmospheric soles (aerosol, hydrosol) and plasmosols, which are involved in the phase transitions of atmospheric water by triggering the removal of metastable states of water into stable ones by heterogeneous nucleation.

For the purpose of this thesis and the work that will follow in the next chapters, we will stick the study of the parameters mentioned above. Despite the great interest that cloud classification represents; dynamics, and the astounding beauty born of the mixing of both, our modelling perspective will prevent us to consider cloud by other means than liquid water content (LWC), ice water content (IWC), optical depth, particle number concentration and cloud fraction.

Before diving deeper into the heart of this thesis' topic, the reader is encouraged–now and at any moment–to put aside its reading and reach the nearest window in order to take a look at the clouds. With some patience and attention, they might have the chance to spot and recognise one of the particular cloud phenomena illustrated on Fig. I.8, and exclaim: "Quanto sono belli!".

I.3.2 Cloud role in climate

As already mentioned, clouds cover approximatively two thirds of the earth's surface. Arising from the evaporation of water, they are more present above oceans than continents. Some marine areas, where low pressures are semi-permanent (North Pacific and Atlantic), show an average cloud cover of 90 %. Clouds have two main roles in the climate system. First, they are at the heart of the water cycle. They greatly participate to the water transport, and the processes that govern their formation and evolution are therefore responsible for the majority of the repartitions of biomes. They also transport energy through the latent heat of water vapour. As water needs energy to evaporate, it stores this energy when in gas phase. The associated enthalpy of vaporisation, expressed in energy unit per unit of mass, is of $2.3e6\,\mathrm{J\cdot kg^{-1}}$ for the water. When the vapour condenses into cloud droplets, it gives back this energy to the air which induces a local warming. From there, air convection can be induced, sometimes leading to more condensation and more energy release. This makes them one of the most sensitive object of the climate system (**bony_clouds_2015**). Super cloud formations, like thunder storm clouds or cumulonimbus, can deploy $20\,000\,\mathrm{GW}$ (**chalon_combien_2014**). This is a hundred times more than the sum of all active power plants in France. Thus, clouds have the potential for great transformations, and are indeed quasi systematically involved in extreme weather events.

In parallel, clouds stand at the interface between downward and upward radiative fluxes. Their white colour indicates that they reflect efficiently the incoming solar radiation (SW), which also explains why some clouds cast dense shadows on the ground. On the other hand, the water vapour they contain is a very efficient greenhouse gas. But absorption of terrestrial radiation in clouds is dominated by condensed liquid water droplets, which makes clouds very effective at absorbing upward infrared fluxes (LW) coming from the surface. The net perturbation of the radiative budget due to those two effects of clouds is called the Cloud Radiative Effect (CRE). It depends on both environmental factors and cloud properties:

- **Cloud properties**

./chap1/images/compile_nuages_v2.png

Figure I.8: Photographies of different cloud types and phenomena. Cumulus congestus over Sorbonne

  - Optical thickness (τ): controls how much short wave is scattered/absorbed and how opaque the cloud is for long wave.
  - Cloud-top altitude: drives the cloud-top temperature which itself determines the upward long wave emissions. Higher clouds have stronger long wave warming (i.e. greenhouse effect).
  - Cloud particles effective radius (r_e): at constant water content, cloud particles radius determine the number of particles. Short wave reflectivity is proportional to particles number concentration (N_p), but also varies in r_e^2.
  - Cloud phase: liquid droplets and ice crystals have different optical properties (scattering and emissivity), and thus affect differently the CRE.

- **Environmental factors**
  - Solar zenith angle: cloud apparent thickness varies with the incident angle of short wave radiations. This is linked to the time of day: during the night, clouds have by definition no short wave effect, only long wave. This is particularly important in polar regions where daytime highly varies.
  - Surface albedo: over bright surfaces, clouds will not increase much the apparent albedo, leading to a smaller short wave effect at the top of the atmosphere.
  - Aerosol particles: impact cloud formation, lifetime, precipitation, particles number and phase (see Sect. I.3.3 and Sect. I.4.1).

The CRE is not necessarily in balance for a singular cloud. On the contrary, it is most of the time negative at the surface. In other words, in spatial and temporal average, clouds usually cool the surface. This is true at all latitudes except in polar regions where the clouds CRE at the surface is positive (see Fig. I.21).

I.3.3 Cloud formation and evolution

Clouds are among those physical systems that involve a wide range of disciplines. Cloud physics is a subtle combination of thermodynamics, fluid mechanics, radiative physics, and molecular physics. In this section, we will explore some of the main mechanisms that allow the existence of clouds in Earth's atmosphere.

I.3.3.1 Notions of stability

The atmosphere–or more precisely, a parcel or a layer of it–can be classified into three different stability states. These characterize whether the considered atmospheric layer is convective, stable, or neutral. Let us first consider the case of *dry air*.

The stability of a dry air parcel depends on the vertical balance between its weight and the buoyancy force:

$$m_p \vec{a} = \vec{\Pi} + \vec{P}_p \tag{I.1}$$

where m_p is the mass of the air parcel, $\vec{a}$ its acceleration, $\vec{\Pi}$ the buoyancy force, and $\vec{P}_p$ the weight of the parcel. The buoyancy Π depends on the density ρ_e of the surrounding air and the density of the parcel ρ_p. An fluid parcel less dense than its surrounding fluid floats in it; conversely it sinks.

The equation of state for air, considered as an ideal gas, is

$$P = \rho R_a T \tag{I.2}$$

where R_a is the gas constant for dry air, equal to $287\,\mathrm{m}^2 \cdot \mathrm{s}^{-2} \cdot \mathrm{kg}^{-1}$. Thus,

$$\rho = \frac{P}{R_a T}. \tag{I.3}$$

Since the pressure of the parcel and its surroundings are identical, it is the temperature difference between T_p (the parcel temperature) and T_e (the temperature of the surrounding environment) that determines the ratio $\frac{\rho_p}{\rho_e}$ and, therefore, the stability state of the atmospheric layer.

Let us consider an atmosphere initially in hydrostatic equilibrium (i.e. in balance between the pressure gradient force and gravity):

$$-\frac{1}{\rho_e}\frac{\partial P}{\partial z} = g \tag{I.4}$$

A change in density from $\rho_e \rightarrow \rho_p$ induces a vertical acceleration a of the air parcel (now with a density ρ_p different from the environment):

$$-\frac{1}{\rho_p}\frac{\partial P}{\partial z} = g + a \tag{I.5}$$

Substituting eq. (I.4) into this expression, and using the equation of state, yields:

$$a = -g\frac{T_e - T_p}{T_e} \tag{I.6}$$

Equation (I.6) can be rewritten as a differential equation by assuming linear temperature profiles of the form:

$$T_i = T_0 + z\,\Gamma_i \qquad (i = e, p) \tag{I.7}$$

where Γ is the vertical temperature gradient, or *lapse rate*. The vertical derivative of potential temperature is:

$$\frac{d^2z}{dt^2} = -\frac{g}{T_0}\frac{d\theta}{dz}z \tag{I.8}$$

The term $\frac{g}{T_0}\frac{d\theta}{dz}$ is the squared Brunt–Väisälä frequency[12]:

$$N^2 = \frac{g}{T_0}\frac{d\theta}{dz} \tag{I.9}$$

The Brunt–Väisälä frequency characterizes the oscillations of a fluid parcel that has been displaced, by any perturbation, from its hydrostatic equilibrium position. When the balance between weight and buoyancy is disturbed, these forces act to restore the parcel to equilibrium. Like a spring given an impulse, the parcel will oscillate at frequency N around its equilibrium position, if the oscillations are maintained. Such oscillations occur both in the atmosphere and in the ocean, where they are known as *gravity waves*.

It is often convenient to visualize the dry adiabatic stability as follows:

Stable case: When the lapse rate of the parcel is steeper (i.e., more negative) than that of the environment ($\Gamma_p < \Gamma_e$, with $\Gamma_e = -g/c_p$), the atmosphere is stable. If the parcel is given an upward motion, buoyancy acts as a restoring force, returning it to its equilibrium altitude. This situation is analogous to a mechanical potential well associated with a stable equilibrium.

Unstable case: When the lapse rate of the parcel is weaker (less negative) than that of the environment ($\Gamma_p > \Gamma_e$), the atmosphere is unstable. In this case, if the parcel is given kinetic energy that displaces it from equilibrium, buoyancy amplifies its upward motion. This is analogous to a mechanical potential peak associated with an unstable equilibrium.

Neutral case: When the lapse rates are equal ($\Gamma_p = \Gamma_e$), the atmosphere is neutral. Once the parcel's kinetic energy is dissipated, buoyancy and gravity are in balance, keeping the parcel at its new altitude.

However, in process of cloud formation, the air parcel is not in dry but saturated conditions. The temperature gradient of the air parcel is not any more Γ_d (dry) but Γ_s (saturated). The expression for the latter is derived from the mathematical expression of the first law of thermodynamics, accounting for latent heat release, and assuming hydrostatic balance:

$$dq = c_p dT + g dz + L_v dq_s \tag{I.10}$$

with dq the variation of internal energy, c_p the specific capacity of dry air, L_v the latent heat of vaporisation, and dq_s the variation of saturation specific humidity. Assuming an adiabatic

[12] Actually, N has the dimension of a pulsation.

./chap1/images/stability.png

Figure I.9: Atmospheric stability of dry air by temperature gradient (B), and associated mechanical analogy (A).

transformation (i.e. $dq = 0$), and dividing by $c_p dz$, we get,

$$\frac{dT}{dz} = -\frac{g}{c_p} - \frac{L_v dq_s}{c_p dz} \tag{I.11}$$

The term $\frac{g}{c_p}$ is the dry adiabatic lapse rate Γ_d (or $\left(\frac{dT}{dz}\right)_d$ when the latent heat is not considered). By writing the change in q_s as a function of temperature rather than altitude,

$$\frac{dT}{dz} = -\frac{L_v}{c_p}\frac{dq_s}{dT}\frac{dT}{dz} - \Gamma_d \tag{I.12}$$

and by rearranging the terms, we get the expression of the temperature gradient of saturated ascent, or *saturated adiabatic lapse rate*:

$$\Gamma_s = -\left(\frac{dT}{dz}\right)_s = \Gamma_d \left(1 + \frac{L_v}{c_p}\frac{dq_s}{dT}\right)^{-1} \tag{I.13}$$

Because of the release of latent heat, Γ_s is always lower than Γ_d. The stability conditions now expand to five cases: absolutely stable, saturated neutral, conditionally unstable, dry neutral and absolutely unstable.

Table I.1: Stability criteria of moist air.

Criterion	**Case description**
$\Gamma_e < \Gamma_s$	Absolutely stable
$\Gamma_e = \Gamma_s$	Saturated neutral
$\Gamma_s < \Gamma_e < \Gamma_d$	Conditionally unstable
$\Gamma_e = \Gamma_d$	Dry neutral
$\Gamma_e > \Gamma_d$	Absolutely unstable

When an air parcel is lifted by unstable conditions, the phenomenon is called *free convection*. Conversely, *forced convection* refers to upward air displacement caused by a pressure gradient or by topography. The lifted parcel may meet thermodynamically favorable conditions for its vapour to condensate. In case of low tropospheric clouds like cumulus (see Fig. I.8), the *sine qua none* condition for droplet formation is often on temperature. When the isotherm that allows the cloud to form is reached, the vapour condensates and the cloud forms. That explains why those kind of clouds have very flat and homogeneous bases. The air parcel can eventually reach an altitude where it becomes stable, and thus stop its ascension. In a stable case, the cloud can be subjected to gravity waves.

In a saturated ascent, when the droplets form, the latent heat associated with the original evaporation of vapour is released, which warms the surrounding air. This tends to fuel the air parcel lifting. If the total latent heat of the parcel is sufficiently important, and the negative potential temperature gradient is strong enough, the lifting can enter in a positive feedback loop that causes it to self sustain until all the vapour has condensed. We talk of *deep convection*. In such situations, the cloud can extend up to the tropopause where the tropospheric temperature gradient gets cancelled, and so does the buoyancy of the air parcel. This mechanism is at the origin of cumulonimbus formation (Fig. I.8).

I.3.3.2 Cloud phase

On Earth, tropospheric clouds are mostly composed of water, which is present under its three phases in the atmosphere: liquid, solid, and gaseous. Clouds can be classified as super-cooled liquid clouds (SLCs), solid-phase clouds (SPCs), or mixed-phase clouds (MPCs), in which supercooled water droplets and ice crystals coexist.

When the formation of one or several of these phases occurs without the intervention of a pre-existing surface (i.e. aerosols), it is referred to as *homogeneous nucleation* (c.f. Sect. I.3.4). The formation of water droplets by homogeneous nucleation can only occur at supersaturation levels of several hundred percent, which are never encountered on Earth. Likewise, only below $-38\,°C$ can ice crystals form through a homogeneous process (**seinfeld_atmospheric_1998**). Homogeneous solid-phase nucleation, or *ice nucleation*, is observed almost exclusively during the formation of cirrus clouds (high tropospheric clouds).

It is because their formation is facilitated by the presence of aerosols–liquid or solid particles suspended in the air (c.f. Sect. I.4)–that clouds still cover more than two-thirds of the globe's surface. When aerosol particles help the formation of cloud particles, the process is called *heterogeneous nucleation*. This process drastically lowers the humidity level required for liquid-phase formation, and increases the temperature at which ice crystals form; as a result, either phase can develop from small supersaturations, and ice crystal formation can begin at temperatures as high as $0\,°C$.

I.3.3.3 Precipitations

Clouds are known to be at the origin of rain and snow, but not all clouds precipitate. Some clouds do not precipitate significantly, like cumulus humilis and stratus, whilst others are intrinsically precipitating clouds, like nimbus, the *rain clouds*. Cirrus are also precipitating clouds, but their icy precipitations never reach the ground. Deep convective clouds, that are born as cumulus and might end up as cumulonimbus, can precipitate both liquid water and agglomerates of ice crystals under the form of hail or graupel.

Since clouds are composed of liquid or solid water particles in suspension, a balance exist between their sustaining force and the pull of gravity. Precipitation happens when the mass of cloud particles becomes large enough to overcome the ascendant motion of air. Therefore, for precipitations to happen, cloud droplets or ice crystals have to grow, which can happen through several processes (table I.2).

A precipitation event does not necessarily signify the end of a cloud, but can favour it. For a cloud to vanish, the evaporation and/or precipitation rates have to become greater than the cloud particles formation rate (condensation or nucleation) during a time long enough for all the cloud water to leave the suspended solid or liquid phase.

I.3.4 Physics of ice nucleation

We saw that ice nucleation can happen in a pristine atmosphere through what is called homogeneous nucleation, or at higher temperatures with the help of aerosols particles through heterogeneous nucleation. Once cloud temperatures fall below $0\,°C$, ice crystals can form within it. This process, called ice nucleation, denotes both the phase transition of vapour to ice, and

Table I.2: Microphysical growth processes

Growth type	Processes	Description
Diffusional growth	Condensation (liquid)	Water vapour condenses onto liquid droplets.
	Deposition (solid)	Water vapour deposits directly onto ice crystals.
	Wegener–Bergeron–Findeisen (WBF) process	Ice grows at the expense of evaporating supercooled droplets.
Accretion growth	Collision–coalescence (liquid)	Droplets collide and merge into larger drops.
	Aggregation (solid)	Ice crystals stick together upon collision.
	Rimming (liquid to solid)	Supercooled droplets freeze upon contact with ice.
Multiplication growth	Secondary formation	Formation of new particles from fragments or splinters.

the freezing of liquid droplets. Unlike the formation of cloud droplets, which never occur homogeneously (i.e. without the help of an aerosol particle) in the atmosphere, the freezing of cloud particles can happen by both homogeneous and heterogeneous nucleation. Since we focus on aerosol-cloud interactions (ACI) in this thesis, we are particularly interested in the second process. Furthermore, the fact that ice crystals can form homogeneously and heterogeneously gives to ice nucleation an additional dimension: aerosols particles that can nucleate ice have the ability to completely change a cloud's life story. This is why heterogeneous ice nucleation is particularly interesting and crucial when modelling clouds; uncertainties on heterogeneous ice nucleation are at the root of cloud–and by extension climate–modelling uncertainties.

I.3.4.1 The energy barrier

In order to understand the physics behind heterogeneous nucleation, it is necessary to recall how homogeneous nucleation works. The following are the basics of classical nucleation theory (CNT). What prevents the spontaneous formation of an ice crystal from water vapour of a liquid droplet, is the competition between two energy terms: a volume term favorable to the phase transition, and a surface term that resists the nucleation.

$$\Delta G = \Delta G_{\text{vol}} + \Delta G_{\text{surf}} \tag{I.14}$$

Let us assume the formation of a spherical ice embryo of radius r. The volume term comes from the released latent heat of the phase transition, which lowers the Gibbs free energy of the system. It scales with the volume of the ice embryo:

$$\Delta G_{\text{vol}} = \frac{4}{3}\pi r^3 \Delta g_v \tag{I.15}$$

Δg_v is the volumetric Gibbs free-energy change associated with the phase transition, and expressed in Jm^{-3}.

The surface term expresses the energy cost due to the creation of an ice-water or ice-vapour interface. It is due to the molecules unsatisfied bonds at the phases interface, and is the origin of surface tension. The term scales with the surface area:

$$\Delta G_{\text{surf}} = 4\pi r^2 \sigma_i \tag{I.16}$$

where σ_i is the surface tension between the new phase and the parent phase: $\sigma_{i,v}$ for vapour to ice, and $\sigma_{i,l}$ for liquid to ice, both assumed to be anisotropic.

The Gibbs free energy change of nucleation thus express as:

$$\Delta G(r) = \frac{4}{3}\pi r^3 \Delta g_v + 4\pi r^2 \sigma_i \tag{I.17}$$

The surface term $\propto r^2$ opposes formation; volume term $\propto r^3$ favours growth when $|\Delta g_v|$ is large enough. The maximum of $\Delta G(r)$ is the nucleation energy barrier ΔG_c at critical radius r_c. They are derived for $\frac{\partial \Delta G(r)}{\partial r} = 0$, which lead to the critical radius expression:

$$r_c = -\frac{2\sigma_i}{\Delta g_v} \tag{I.18}$$

Injected in the expression of $\Delta G(r)$, we get:

$$\Delta G_c = \frac{16}{3}\pi \frac{\sigma_i^3}{|\Delta g_v|^2} \tag{I.19}$$

where ΔG_c is in J, σ_i in $\mathrm{J \cdot m^{-2}}$, Δg_v in $\mathrm{J \cdot m^{-3}}$ and r_c in m.

The nucleation rate is usually written as follows,

$$J = J_0 \, \exp\left(-\frac{\Delta G_c}{k_B T}\right) \tag{I.20}$$

expressed in s^{-1} and where J_0 is a kinetic prefactor, and k_B the Boltzmann's constant.

The expression of Δg_v varies whether the phase transition is vapour to ice, or liquid to ice:

Vapour to ice: In this case, the driving force for the ice embryo formation is the vapour supersaturation relative to ice:

$$S_i = \frac{p_{\text{vapour}}}{p_{\text{sat,ice}}(T)} \tag{I.21}$$

The temperature and k_B intervene, along with v_{molecule} the molecular volume of ice, to express

Δg_v as a volumetric energy density:

$$\Delta g_{v,v} = -\frac{k_B T}{v_{\text{molecule}}} \ln S_i \tag{I.22}$$

When the vapour is supersaturated ($S_i > 1$), $\Delta g_{v,v} < 0$ which favours solid growth in eq. (I.17). The higher the S_i, the lower the critical radius r_c (eq. (I.18)) and the energy barrier ΔG_c (eq. (I.19)).

Liquid to ice: Here, the driving force for phase transition is the supercooling $\Delta T = T_m - T$. The latent heat released during freezing of liquid water L_f and the ice density ρ_i have to be taken into account in order to express $\Delta g_{v,l}$:

$$\Delta g_{v,l} \approx -\rho_i L_f \, ln \frac{T_m}{T} \tag{I.23}$$

where we make the approximation that $L_f \approx \Delta h$, the enthalpy of changing state. When the supercooling is positive (i.e. the temperature is under the melting point of water T_m), $T_m > T$ which implies $\Delta g_v < 0$. This create favorable condition for ice growth and lowers the critical radius and the energy barrier.

In the case of heterogeneous nucleation, the energy barrier is reduced by the presence of what is called an ice nucleating particle (INP). CNT describes the effect of INPs through the contact angle θ between it and the ice embryo (Fig. I.11 A). Irregularities called *active sites* on the INP surface, such as bumps or crevasses, help the accumulation of water molecules for the formation of an ice embryo. Consequently, ΔG_c is lowered by a factor f that depends, in CNT, on the contact angle:

$$\Delta G_{c,het} = f(\theta)\, \Delta G_{c,hom} \tag{I.24}$$

The presence of an INP can reduce so much the energy barrier, that heterogeneous nucleation can happen at much warmer temperature than homogeneous nucleation (Fig. I.11 B).

Note about the $-38\,°\mathrm{C}$ threshold

It is common to consider the temperature condition for homogeneous nucleation as a threshold in the vicinity of $-38\,°\mathrm{C}$, but the physical origin for that threshold is rarely explained nor even mentioned. However, the elements of CNT developed herein above can allow us to understand it. Figure I.12 shows the ice nucleation rate for homogeneous freezing from liquid droplet to ice (green line). One can see that it is extremely low at temperatures above $-40\,°\mathrm{C}$, and can be considered as zero above $-30\,°\mathrm{C}$. On the other hand, it gets to values that could allow actual formation of ice crystal below $-40\,°\mathrm{C}$, and even higher rates below $-50\,°\mathrm{C}$. The blue line depicts the average mean time of nucleation. It is even more striking that above $-40\,°\mathrm{C}$, the homogeneous nucleation cannot happen. At $-35\,°C$, it takes 10^{18} s for a droplet to freeze, or approximatively 32 billion years. However, at $-50\,°C$, it only takes $10\,s$ on average.

In conclusion, this threshold of $-38\,°C$ ($235\,K$) is half quantitative-half qualitative. It marks the beginning of acceptable and rapidly increasing values of nucleation rate, but still very few ice crystals can form homogeneously at this temperature. Therefore, INPs active at temperature around the homogeneous temperature threshold (see Sect. I.4.3) can still have a significant impact on the solid phase of cloud.

Nota bene: CNT is a powerful tool to understand the physics of nucleation. However, and even though its attempt at describing heterogeneous nucleation is very physically based, it remains a poor description of the complexity of aerosol-cloud interactions happening in a nucleation event. **fletcher_active_1969** have shown that INPs can have complex structures of active sites that can greatly influence the nucleation rate. Many recent studies have tried to take this phenomenon into account, using so-called *active sites* or *time independent* ince nucleation approaches (**murray_ice_2012**; **niemand_particle-surface-area-based_2012**; **kanji_heterogeneous_2013**; **hummel_simulating_2018**).

I.3.4.2 Types of heterogeneous ice nucleation

Heterogeneous ice nucleation can happen through four different processes.

Immersion freezing: Immersion freezing refers to the formation of an ice crystal from a droplet inside which an INP is immersed. It can occur when saturation with respect to ice[13] is greater than unity (i.e. $S_i \geq 1$) and when $T \leq 0$ °C, but requires the preliminary formation of a liquid droplet, and thus experiencing a sufficiently high supersaturation for liquid condensation to have taken place.

Deposition nucleation: Along with immersion freezing, deposition nucleation is the dominant process in the formation of ice clouds in the atmosphere. In this case, water vapour deposits directly on the surface of INPs, when the humidity conditions ($S_i \geq 1$) and temperature ($T \leq 0$ °C) allow it and sufficiently active INP are present.

Condensation freezing: When air approaches saturation with respect to liquid water (S_l), water vapour can condense on the surface of INPs without forming a droplet. If saturation is reached but remains below the threshold for droplet formation, an ice crystal can form. In general, INPs contain soluble components that can eventually act as CCN. If condensation freezing does not occur quickly enough, a droplet may form on the INP. In this case, immersion freezing can still occur later if conditions are met.

[13] Saturated vapour pressure is lower over ice than over liquid water surface. Consequently, saturation with respect to ice (S_i) is always higher than saturation with respect to liquid (S_l) at freezing temperatures.

Contact freezing: The collision of a supercooled water droplet with an INP can lead to the formation of an ice crystal if conditions are met ($T \leq 0$ °C and a preexisting supercooled droplet collides with an active INP). Contact freezing is limited by the collision rate between droplets and INPs, which depends on the INP concentration as well as the size of the particles and droplets.

The modelling of ice nucleation will be broached multiple times in this thesis. Chapter **??** will provide a first overview of this topic (**??**), and **??** will dive deeper into the subject of parametrising ice nucleation.

Opening Questions | Modelling of ice nucleation

- How do the different ice nucleation representations compare in a regional model? (**??**)
- Is it possible to use a general parametrisation for Arctic INPs? (**??**)
- Is the presence of INPs in the Arctic important for the formation of clouds and for the modification of their properties? (**??**)

./chap1/images/matus_cloud_frac.png

Figure I.10: Zonal mean of the cloud fraction for different types of clouds (panel A). Liquid clouds (red line), ice clouds (blue line), mixed-phase clouds (purple line) and multi-layer clouds (green line). Panel B shows the annual (black line), june-august (red line) and december-february (blue line) of the zonal mean cloud fraction for mixed-phase clouds. Figure adapted from **matus_role_2017**

./chap1/images/hetero_nucleation.png

Figure I.11: Heterogeneous ice nucleation. Panel A is a schematic of the formation of an ice embryo in contact of an INP. The contact angle θ and the surface tensions σ_i of the different actors are represented (adapted from **lohmann_introduction_2016**). Panel B is the energy barrier ΔG expressed in thermal units, function of the ice embryo radius, for homogeneous and heterogeneous ice nucleation. The curves maximum is the critical Gibbs free energy that, once reached allows the embryo to grow as an ice crystal.

./chap1/images/homogeneous_nucleation_rate.png

Figure I.12: Representation of theoretical droplet freezing rate by homogeneous nucleation (liquid to ice) versus temperature. The corresponding average time needed for an ice crystal to form is plotted on the second axis. In the vicinity of $-38°C$, the nucleation rate rapidly decreases with increasing temperature, while the average time before freezing dramatically increases.

I.4 Ice nucleating particles in the Arctic

In the previous section, the importance of aerosol particles for the formation of clouds has been made tangible (c.f. Sect. I.3.3.2 and I.3.4, Fig. I.11). Indeed, among the most challenging issues in climate modelling are cloud microphysics, especially in the representation of aerosol-cloud interactions (**bony_clouds_2015**). This includes cloud formation, phase partitioning, precipitations, and concerns all the associated radiative consequences.

Aerosol particles, independently of their nucleation ability, can be defined as non-pure[14] water liquid or solid particles in suspension in the air. They can be emitted at the surface, from an erodible ground, vegetation or from the ocean. In this case we talk of primary aerosols. They are categorised as secondary aerosols when their properties are issued of a physical or chemical transformation of gaseous precursors that happened when they were already in suspension.

The formation of liquid particles by heterogeneous nucleation with cloud condensation nuclei (CCN) is an intense field of research and modelling since it concerns the great majority of clouds on Earth. Nevertheless, the ice phase is particularly important in polar regions, where the majority of clouds contain ice (see Fig. I.10). As mentioned before, this thesis focuses on the processes related to cloud ice formation, consequently, this section will investigate the nature and properties of INPs.

I.4.1 The variety of aerosols effects on cloud and climate

Before getting to the particularities of INPs, let us take a glimpse to the variety of aerosols effects on clouds and climate.

The first effect of aerosols is what is called the *direct effect*, or aerosol-radiation interaction (ARI). It is related to the optical properties of aerosols particles. Indeed, they can reflect, absorb, scatter or transmit incident short wave (and long wave when their sizes get close to the IR wavelengths) radiations, which impact the radiative balance of the atmosphere. Beyond the direct effect, exist two other types of impact of these particles on the climate energy budget. Those concern the particles interactions with clouds (Fig. I.13).

Semi-direct effects: The presence of an aerosol plume, through its interactions with solar radiation, causes a warming of the air layer it occupies. This modification of the vertical temperature profile of the atmosphere induces, when the temperature decreases with altitude, an increase in hydrostatic stability. This phenomenon can inhibit atmospheric convection and consequently alter cloud formation by reducing their vertical development in favour of horizontal spreading. This leads to an increase in stratus cloud coverage and a reduction in cumulus development. In the Arctic, due to the low solar exposure associated with high latitudes, and because of the scarcity of aerosols particles in such remote region, the direct and semi-direct

[14] Aerosol particles can be aqueous, i.e. contain water.

effects of aerosols are not very pronounced. Indirect effects have the most significant impact on the region's energy budget.

Indirect effects: Indirect effects are the consequences of aerosol–cloud interactions. While these interactions can be very complex, the general sequence of mechanisms is straightforward to understand. An increase in CCN concentration will promote the formation of a larger number of droplets, which in turn increases the cloud's optical thickness and infrared emissivity (**twomey_influence_1977**). This is referred to as the first indirect effect. Furthermore, for a constant liquid water content, an increase in droplet number leads to a decrease in their mean size. The coalescence processes responsible for droplet growth will then be less efficient, delaying cloud precipitation. This is the second indirect effect, which results in an increased cloud lifetime (**albrecht_cloud_1989**).

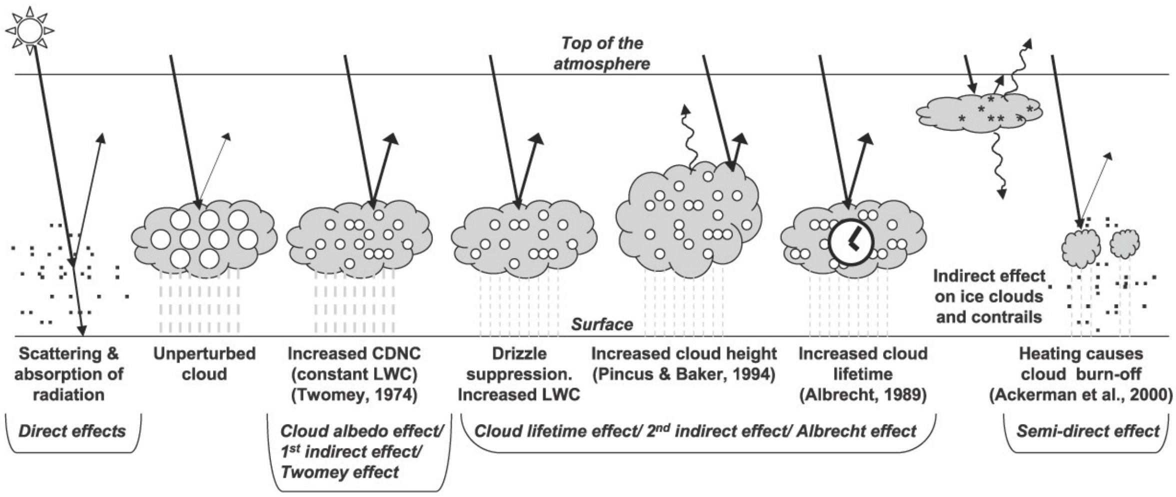

Figure I.13: Schematics of the different main aerosols effects on clouds and the radiative budget. Graphics from **ipcc_ar4_2007**.

Many other indirect effects exist, such as the impact of aerosols on cloud altitude or on ice clouds. These processes, however, remain poorly understood, mainly due to gaps in our understanding of aerosol–cloud interactions (**murray_opinion_2021**).

Figure I.14 shows the estimation of the total radiative forcing due to aerosol particles. It divides into two the categories aerosol-radiation (direct and semi-direct effects) and aerosol-cloud (indirect effects) interactions. The overall contribution of aerosols effects is negative, about $-1.3\,\mathrm{W\cdot m^{-2}}$, with an important uncertainty of $\pm 0.8\,\mathrm{W\cdot m^{-2}}$. The latter is mainly due to aerosol-cloud interactions which constitute a great observational and modelling challenge. On the last line is shown the spreading of CMIP models output for the aerosols forcing. Even though it gets narrower between CMIP5 and CMIP6, it is striking that the uncertainty comes from a profound understanding issue of the implied processes. In comparison, the uncertainty on CO_2 radiatif effect (Fig. I.5 A) is only about $12\,\%$.

./chap1/images/IPCC_AR6_WGI_TS_Figure_15_aerosols.png

Figure I.14: Effective radiative forcing for aerosols (sum of SO_2, organic carbon, black carbon and ammonia), from 1750 to 2019. Effect due to aerosol-cloud interactions (deep purple) is distinguished from aerosol-radiation effect (light purple). The uncertainties have reduced since the fifth IPCC Assessment report, but still remain important. Graphics from **intergovernmental_panel_on_climate_change_climate_2023**.

Furthermore, the assessment of IPCC presented on Fig. I.14 does not account for all the particles that can be implied in aerosol-cloud interactions (ACI). As it will be discussed in the next sections, aerosols from mineral dust and biological matter from oceanic and continental areas, are anticipated as impactful factors for cloud ice phase. The corresponding effect are barely quantified, and their evolution in the warming climate remains unknown. These side effects of ACI are among the core topics addressed in this thesis.

I.4.2 Physical requirements of INPs

For an aerosol particle to be able to act as an INP, its physical properties have to meet specific requirements. Because of the way heterogeneous ice nucleation works, i.e. the formation of a molecule cluster on active sites–which helps the ice embryo to reach the critical size of nucleation––, good INPs are rather insoluble particles. For the same reason, INPs size is a critical factor for their efficiency. Indeed, the number of active sites is proportional to the particle surface area. Its type (mineral, carbon organic or inorganic) will partly determine the active sites density of the INP. The crystallographic structure of the particle also have an important role: the closer it is to ice's, the easier nucleation gets. Eventually, its molecular bindings with water (i.e. is it hydrophilic or hydrophobic) can impact its nucleation abilities.

The INPs are often distinguished by their activation temperature, i.e. the temperature at which they form ice crystals. In the following, we will call warm INPs particles that nucleate ice at temperatures higher than $-15\,°C$, and we will talk about cold INPs when the activation temperature is lower than this arbitrary threshold. In this thesis, we will encounter very warm INPs that can nucleate ice up to $-3\,°C$, but we will rarely mention very cold INPs (efficient at temperatures around the homogeneous nucleation threshold) like soot.

I.4.3 Natural or anthropogenic? The diversity of INPs

Arctic INPs are poorly characterized, but the current state of research suggests that they are mainly primary aerosols: mineral dust and biological particles from marine or terrestrial origin (**creamean_marine_2018**; **hartmann_terrestrial_2021**). For each of these INP types, several impacts of climate warming on their emissions–and therefore on their concentrations in the Arctic atmosphere–can be identified. The following paragraphs review the different types of these particles.

I.4.3.1 Sea salts and marine organics

Sea salts are water-soluble in composition, which makes them poor INPs. However, they play a major role as CCN. Moreover, they share their emission mechanism with marine organic particles, which are very efficient INPs (**gantt_physical_2013**; **burrows_ice_2013**). These two aerosol types are grouped under the term *sea sprays*, which refers to their emission mechanism.

These aerosols are emitted during the bursting of fine water droplets projected at the ocean surface (illustrated on Fig. I.15). The lifting of foam by swell and wind is the main production mechanism (**vignati_global_2010**). Thus, changes in surface winds in oceanic regions influence INP concentrations, but it is primarily the variation of ocean surface area in contact with the atmosphere (open ocean) that determines the emission of marine aerosols. This is relevant in polar regions, especially in the Arctic where the open ocean surface changes with both the seasonal cycle of sea ice and its multi-decadal decline associated with climate warming. This link between sea ice and the emission of sea sprays (sea salts and marine organics) suggests that the ongoing decrease in sea ice concentration–seemingly inevitable given current warming trajectories–will be accompanied by an increase in the concentration of this aerosol type. Let us mention emissions from open leads in the ice pack (see Sect. I.5.1), that follow different mechanisms but can contribute significantly to sea sprays (**lapere_modeling_2024**).

I.4.3.2 Biological sources, terrestrial and marine

Particles produced by biological activity can act as INPs when emitted into the atmosphere, and in some cases—such as bacteria—can be particularly active at temperatures above -15°C (**storelvmo_aerosol_2017**). Pollen, plant debris, bacteria, and viruses are the main representatives of continental organic aerosols. Marine organic INPs consist of residues from biological activity, such as algal, planktonic, or phytoplankton debris.

creamean_ice_2019 show that warming of the Arctic Ocean and the retreat of sea ice cover are two factors that promote biological activity at high northern latitudes. Similarly, **creamean_thawing_2020** describe how permafrost thaw exposes the atmosphere to important INP sources through the re-emergence of biological organisms, once again exposed to the atmosphere. In general, temperature acts—up to a certain point—as a catalyst for microbial activity. Climate warming is therefore expected to mechanically increase the presence of biological INPs in cold regions. For example, it drives the migration of marine biological activity toward higher latitudes at a rate of about 1° per decade (**renaut_northward_2018**). All of these temperature-driven changes could lead to an increase in biological INP concentrations in the Arctic.

I.4.3.3 Mineral dust

Among aerosols capable of acting as INPs, mineral dust is often considered as the most common, and consequently as the most important in mass concentration (**kok_contribution_2021**). This relies on their large size (which confer them high amount of active sites) and on the high emissions fluxes of such particles in the atmosphere (**kok_mineral_2023**). Mineral dust is most often lofted by wind or projected into the air through the impact of other particles (saltation and sandblasting processes). They originate mainly from semi-arid regions or deserts but can also be emitted from riverbeds or glaciers (**meinander_newly_2022**). Again, climate warming is expected to increase the number of sites favorable to the emission of these particles through glacier melt and riverbed drying. In areas where warming leads to a retreat of arid zones due to increasing biodiversity, the decrease in mineral INPs could be partially offset by the rise in biological sources. Although they are active at lower temperature than biological INPs, they can participate to transport the latter and thus extend their range of action. However, due to their relatively short atmospheric lifetime—linked in part to their high density–they can also limit the transport of biological INPs.

I.4.4 Lifetime and deposition processes

Unlike some stable chemical species such as CO_2, aerosols have a short lifetime in the atmosphere. Several processes contribute to the decrease in atmospheric aerosol concentrations, so that their residence time in the troposphere ranges from a few hours to about ten days. These processes can be classified into three types: dynamical, chemical, and microphysical.

In the absence of any interaction with other chemical elements, the lifetime of aerosols in the atmosphere is limited by dynamical processes. The main ones are the mechanisms of dry and wet deposition. Dry deposition refers to the sedimentation of particles under the effect of gravity and Brownian motion. It depends on the balance between the weight of the aerosols and the various resistances to this sedimentation (**farmer_dry_2021**). However, dry deposition accounts for only $10\text{-}20\,\%$ of aerosol removal from the atmosphere; wet deposition is the dominant dynamical process. Near emission sources, on the other hand, various dry deposition processes dominate, particularly those affecting larger particles. One example is impaction: when the inertia due to the particle's mass ejects it from the trajectory of the air parcel carrying it, possibly leading to its collision with the surface. Similarly, interception refers to the collision of the particle with the surface due to its size when its trajectory passes too close to it. Smaller particles are more affected by Brownian motion, which is the dominant process in their dry deposition.

The term wet deposition refers to the removal of aerosols from the atmosphere by precipitation. This can occur in different ways. Aerosols can be carried to the ground by falling precipitation (rain or snow) when they are located below or within a cloud: this is the *washout* process (or impaction scavenging). Alternatively, after serving as cloud condensation nuclei or ice nuclei, aerosols are trapped in droplets or ice crystals and are brought to the ground during the precipitation of the cloud particles they helped forming. This is referred to as *rainout* (or nucleation scavenging).

As will be seen in the next chapters, these processes must be taken into account when attempting to simulate the transport of aerosols in order to trace them back to their emission sources (**??**), and when simulating aerosols particles emission and transport (**??**).

I.4.5 Seasonal variability and observational ability

Like many variables and atmospheric species, INPs can present a seasonal cycle that could eventually be detected in the observed concentrations. Theoretical explanations for this variability appear to be as a matter of course. Emissions can be subject to weather conditions (wind, snow cover, temperature), to vegetation cycle (for continental organic INPs) or to sea ice cover (for oceanic emissions). Atmospheric transport of particles is also subject to synoptic weather configurations that vary along the year, which can affect the local concentrations of remote sourced species. Furthermore, the wind and precipitation can have a strong impact on particles removal (c.f. Sect. I.4.4).

Figure I.16 illustrates how much INP concentrations can vary across the seasons. Panel A shows the concentrations for five activation temperatures, ranging from $-10\,°\mathrm{C}$ to $-18\,°\mathrm{C}$, of INP measurements performed at Utqiaġvik (Alaska) over the year 2012-2013 (**wex_annual_2019**). The $-10\,°\mathrm{C}$ series, or concentrations of the warmest INPs, is the one that shows the highest variability. It peaks during summer and spring, around 10^{-2} $\mathrm{INP/L}$, its lowest values are seen during winter down to three orders of magnitude lower. Comparison with panel B, that shows the Arctic average temperature and precipitation, suggests a correlation between the climate seasonal cycle and the observed variability in INP measurements. Without further analysis, the evidence remains light, but the theoretical background mentioned above, still is plausible and suggest a causal relationship. However, this plot suggests another explanation for this apparent variability.

The concentrations for the different activation temperatures show seasonal variability that weaken continuously with the temperature. While the stronger variability is met for the INP active at $-10\,°\mathrm{C}$, concentrations are almost stable at $-18\,°\mathrm{C}$. Whilst this could be explained by the variability inherent to the different species implied on this INP temperature spectrum, the continuous weakening of the variability, and the fact it happens in winter, suggest something else.

Warmer INPs are much more easily activated in clouds during winter when the temperature are lower, while the colder INPs are not as much activated. Thus, the natural in-cloud activation of the warmest INPs would prevent them to get captured by the air inlet of the aerosol filtering instruments. In the meantime, colder INPs would remain inactivated rather because of temperature condition, or because the solid phase would have already been formed on the warm INPs. This would lead to a systematic bias on the measurement of INPs; the warmest being under-represented at the sampling time, and presenting a strong apparent seasonal cycle compared to the steadier cold INP concentrations.

During the one year Arctic campaign MOSAiC (see **??**), **creamean_annual_2022** observe the same variability on INP concentrations: high concentrations during summer and spring, and low to zero warm INPs during winter, while the cold INPs have an almost constant concentration level all the year. It questions the reliability of INP measurement techniques, as well as the conclusions drawn from them. Although the theory behind the natural variability of INP concentrations is solid, it has to be investigated with different techniques than just in situ measurements. In this context, modelling appears to be a powerful tool to test conceptual understandings, to remove doubt and confirm hypothesis. Chapter **??** will investigate this question by addressing the problem of INP modelling.

Opening Questions | Ice nucleation particles

- Which tools are required for the identification of aerosols particles emission sources? (**??** and **??**)
- What are the emission sources of Arctic INPs? (**??**)
- What are the important aerosol species for the ice nucleation of Arctic clouds? (**??**)
- Is it possible to faithfully model INPs in the Arctic with our current knowledge, and how to do it? (**??**)

./chap1/images/nuages/seasprays_pic_HD.jpg

Figure I.15: Sea sprays are emitted during the bursting of fine water droplets projected by wind and waves. They contain sea salts and organic matter from marine life.

./chap1/images/Ninp_cycle_Utqiagvik.png

Figure I.16: One year (2012-2013) time series of INP concentrations active at different temperatures. Panel A shows the raw concentrations alongside polynomial fits. The magnitude of the seasonal cycle varies with the activation temperature. Panel B is the averaged Arctic temperature and precipitation on the same period. Data from **wex_annual_2019**.

I.5 The Arctic atmosphere

The name Arctic finds its roots in ancient Greek: αρκτς [árktos], which means bear. Despite the fact the Arctic is commonly known as the place of polar bears, antiquity Greeks did not know it back then. The exploration of the Arctic region by Europeans is much more recent with many exploratory expeditions in the 19th century. As for the geographical North pole, it has been officially reached only during the advent of flying machines in the 20th century. Theses difficulties in Arctic exploration give evidence for how extreme is its climate, which still make modern observations difficult to carry. Thus, the Greeks were rather referring to the constellations: the Great and the Little bears which embody the north in the sky[15].

The Arctic region could be defined with different geographic thresholds. An astronomical threshold is used to set the latitude of the Arctic Circle, located at 66.5°N. Northward of this line, the current Earth obliquity causes the Sun not to rise on the day of the winter solstice; on the summer solstice, it does not set. The July two meters mean temperature isotherm, is sometimes given as a more meteorological and dynamic definition. Indeed, with the global warming going on (c.f. Sect. I.2.2 and Sect. I.5.2), the Arctic is condemned to shrink. Similarly, the boreal tree line can be used as the Arctic border, but again, with the climate warming, this definition leads the Arctic into shrinking, although in this case, the Arctic Circle will always remains a backup. Figure I.17 gives an insight of the Arctic geography, and illustrates the above definitions.

In this section, we will explore what specificities make the Arctic a particular region in atmospheric and climate sciences.

I.5.1 A glimpse on Arctic climate conditions

As mentioned herein above, the Arctic is by definition a region where the weather is cold and the diurnal cycle very evolutive along the year. This has several implications for the Arctic atmosphere and radiative budget. The so-called *polar night* gives a strong positive surface radiative forcing to clouds. Indeed, because there is no short wave radiations, remains only the greenhouse effect of clouds. Blocking infrared radiations, they strongly warm the surface. The low solar exposure and cold temperature, also causes a strong stability of the atmospheric boundary layer. This leads to very low convection and to keep atmospheric pollutants close to the surface.

Unlike Antarctica which is a continent, the Arctic region is characterised by its ocean. The Arctic Ocean opens on both the North Atlantic and North Pacific. In 2025, it is still covered all year long by sea ice, but the ice pack extent decreases rapidly in area and thickness due to global warming, which has implications in energy feedback loops (see Sect. I.5.2). In terms of radiation, the sea ice acts like a reflective shield, especially when covered by snow. Therefore, the Arctic clouds have lesser cooling effect since they do not increase the albedo when flying over sea ice.

[15]The Antarctic is therefore the anti-bear place, which is twice as true since it is located around the South pole, and because there is no bear there.

./chap1/images/arctic_map.png

Figure I.17: Map of the Arctic, illustrating the different definitions of the region frontiers and its main biomes. The dashed and solid blue lines are the ice edge of respectively the annual minimum and maximum median sea ice extent. The map is adapted from **maillard_boundary_2022**.

The sea ice cover also inhibits the sea spray aerosol emissions of the Arctic ocean, which makes more pristine the air of central Arctic. But mechanisms such as blowing snow[16], or oceanic emissions through open leads[17] can dispute this assertion. On the other hand, open ocean region of the Arctic are known to be home of large plankton and phytoplankton populations, which can be a great source of biological aerosol particles. Independently of the aerosols, the sea ice pack caps the energy the $-2\,°\mathrm{C}$ ocean thermostat. Open leads tend to produce turbulent fluxes that alter the boundary layer stability and thereby participate to cloud formation.

Even though the Arctic is a remote region, it is far less isolated from human activities than the Antarctic continent. Indeed, because of geographical causes, almost $90\,\%$ of the human population live in the boreal hemisphere. For the Arctic climate, it implies–among other things–an increased sensitivity to air pollution. The Arctic haze, which refers to the accumulation of anthropogenic aerosols transported from mid-latitudes into the Arctic during late winter and spring, consists mainly of sulphate, black carbon, and organic matter originating from industrial and biomass-burning emissions. Because of the stable Arctic atmosphere and limited wet scavenging in winter, these particles can persist for weeks and spread over large regions. The Arctic haze affects the regional radiation balance by scattering and absorbing solar radiation and by altering

[16] Blowing snow is the process of uplifted snow by wind, which could lead to the re-emission of ice crystals and aerosols particles into the atmosphere.

[17] Open leads refer to cracks in the sea ice field which open on the ocean.

cloud microphysical properties. Once deposited, those particles can darken the snow and ice, contributing to lower the surface albedo and a local warming.

I.5.2 Polar amplification

The increase in average temperatures associated with global warming is not homogeneous across the globe. In some regions, warming is inhibited (for example, in the Southern Ocean, where the Antarctic Circumpolar Current benefits from the thermal inertia of deep ocean circulation to isolate Antarctica from warming for several decades), while in others it is amplified. As previously mentioned, the Arctic is the region where warming is most pronounced, with a rate of temperature increase about four times higher than the global average (**rantanen_arctic_2022**). Figure I.18 illustrates this meridional inhomogeneity of climate warming. This intense warming has striking consequences, for instance on the sea ice pack: the eighteen last years have seen the eighteen lowest sea ice extents since the beginning of observation 47 years ago. In this section, we review the main mechanisms that can explain this Arctic amplification.

I.5.2.1 Convergence of horizontal heat fluxes

Synoptic horizontal heat transport contributes to the amplification of warming at high northern latitudes. Although the list of relevant processes remains debated (**serreze_processes_2011**), the Atlantic Multi-decadal Oscillation (AMO) appears to play a role in modulating Arctic temperatures. **chylek_arctic_2009** showed that Arctic air temperature variations are linked to AMO phases. After at least sixty years of warming (1880–1940), the Arctic experienced a cooling period associated with a negative AMO phase. Following a loss of about 1°C over thirty years, temperatures began rising again when the AMO shifted back to a positive phase. However, establishing a robust correlation remains challenging, as AMO records only cover about one and a half oscillation cycles.

The Arctic Oscillation (AO) also strongly influences the region's climate (**quadrelli_simplified_2004**). The AO refers to a sea-level pressure (SLP) tripole at mid- and high northern latitudes (Fig. I.19). A positive AO phase corresponds to a low-pressure anomaly centred over the Arctic, which strengthens the jet stream and supports the persistence of the polar vortex. In this phase, polar air remains confined to the Arctic and isolated from warm lower-latitude air. Conversely, a negative AO phase is characterized by an anticyclonic SLP anomaly over the Arctic, weakening the polar vortex. Cold air can then spill into the mid-latitudes, while the Arctic receives intrusions of warm air. Although Arctic warming follows multi-decadal cycles (e.g., AMO influence), it is systematically faster than the global mean. We can thus distinguish two categories of mechanisms: (1) large-scale, multi-decadal drivers influencing the Arctic through climate cycles, and (2) seasonal to regional feedbacks specific to the Arctic. Although the AO is among the phenomena that explain the Arctic climate, the Arctic amplification could participate to attenuate it.

./chap1/images/arctic_ampli.png

Figure I.18: Annual mean surface temperature anomalies for the period 2004-2024, relative to the 1951–1980 mean. The zonal mean anomalies are shown in the lower right panel. Warming reaches up to 3 °C in the Arctic, approximately four times the magnitude observed at the equator. Data from https://data.giss.nasa.gov/gistemp/maps/NASAGISS

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I.5.2.2 Albedo feedback

Among its many specificities, the Arctic Ocean stands out from other basins due to the variability of its albedo. The presence of sea ice increases the basin's albedo to 0.5-0.7 (50-70 % of incoming solar radiation reflected), compared to only 0.06 for open water.

Sea ice thus acts as a reflective shield against solar radiation. Under climate warming, its extent tends to decrease year after year. The surface of open water–i.e., not covered by ice––increases, absorbing more energy (Fig. I.20 A). This temperature rises, directly affecting sea ice formation, making it thinner and less extensive. In parallel, as Arctic Ocean temperatures exceed air temperatures in winter, the excess sensible heat stored during summer is released back into the atmosphere in winter. This delayed effect of the sea ice–albedo feedback explains why temperature anomalies peak in winter (**serreze_processes_2011**).

A similar positive albedo feedback exists over Arctic land surfaces, but involves snow cover only (not sea ice). Snow has a very high albedo unless darkened by soot, ash, or other deposited

Figure I.19: Climatology of sea level pressure (SLP) for the period 1979–2021, averaged over 50–90 °N. On average, the Arctic exhibits a high-pressure anomaly, consistent with a negative phase of the Arctic Oscillation (AO). Data from **copernicus_climate_change_service_era5_2017**.

particles (c.f. Sect. I.4.4). Earlier snow melt–triggered by positive climate forcing (e.g., enhanced greenhouse effect)–reduces albedo and further accelerates melting.

graversen_polar_2009 showed, by simulating Arctic amplification with fixed albedo, that albedo contributed about 15% to the observed amplification. Other processes must therefore be considered to fully characterize Arctic amplification.

I.5.2.3 Lapse-rate feedback

Arctic temperature conditions and low solar exposure impose a strong atmospheric stability, often associated with a temperature inversion in the Arctic boundary layer. The maximum of the vertical temperature profile is not at the surface, but at an altitude ranging from a few hundred meters to more than one kilometre during the Siberian winter. The frequency, intensity, and height of these inversions vary strongly with season and location (**maillard_boundary_2022**). As a result, the excess heat from surface warming is not efficiently transported to higher atmospheric layers, but remains confined near the surface (Fig. I.20 B). Long wave radiation associated with this additional warming is also partly trapped in the lower atmosphere, contributing to positive feedback. **pithan_arctic_2014** identified this mechanism as likely the main driver of Arctic amplification.

I.5.2.4 Planck feedback

The *Planck feedback* refers to a fundamental negative feedback of the climate system: as surface and atmospheric temperatures rise, the earth emits more outgoing long wave radiation, following the Stefan-Boltzmann law. This increased emission acts to counterbalance the initial warming (c.f. Sect. I.2.3). In the Arctic, the effectiveness of the Planck feedback is reduced because surface and lower-tropospheric temperatures are already very low. At such cold temperatures, the rate of change with temperature of black body emission is weaker than at warmer latitudes. This is due to the non-linear dependence in temperature of the Stefan-Boltzmann law. To recall, it expressed the radiative flux F of a black body at temperature T as

$$F = \sigma T^4 \tag{I.25}$$

with σ the Stefan-Boltzmann constant. Thereby, $\partial F/\partial T$ is also non-linear with temperature. At lower temperatures, the derivative is thus smaller, implying that a unit warming in the Arctic leads to a weaker increase in outgoing long wave radiation than in the tropics. Consequently, the stabilising Planck feedback operates less efficiently, contributing to exacerbate the Arctic warming.

I.5.2.5 Effect of polar clouds

Polar clouds are primarily low-level stratus (**shupe_clouds_2011**). During daytime, they play a dual role: reflecting incoming solar radiation and acting as a greenhouse "lid" for terrestrial infrared radiation. Observations by **intrieri_annual_2002** show that the radiative balance of polar clouds is generally positive–warming the lower atmosphere–, except for a short summer period when the surface albedo is minimal and the cloud are thick enough (**maillard_characterisation_2021**). On an annual average, polar clouds have a net warming effect on the surface (see Fig. I.21 A). While this cloud type tends to impose a net cooling at low and mid-latitudes, the Arctic warming effect is explained by the long polar night characteristic of high latitudes. During this period, cloud albedo has no impact on solar radiation, while their greenhouse effect persists and becomes the sole radiative forcing from clouds throughout the polar night. Nevertheless, polar clouds present a negative surface CRE during summer (**matus_role_2017**). This effect is partly due the summer polar day, and to the reduced sea ice extent showed during this period, getting Arctic conditions closer to those encountered at lower latitudes. Furthermore, increasing temperatures lead to a predominance of the liquid phase over the ice phase in clouds. These liquid-rich clouds become optically thicker and have longer lifetimes. The resulting effect is warming, making this process another positive feedback.

I.5.3 Arctic clouds

The Arctic is a wide region with various types of biomes and local climates (c.f. Fig. I.17). Thus, its cloud variety can happen to be large. In this section, we will focus on the clouds of the high Arctic, which means over or next to the Arctic ocean.

As mentioned before (c.f. Sect. I.5.1), the Arctic atmosphere shows a remarkable stability, especially during the polar night. This causes the absence of strong convective system during a large part of the year, which result of a general cover of low clouds, that can often reach the ground as fogs do. The temperature conditions over the Arctic ocean range between $-50\,°C$ in winter, and positive values in summer. Although the ocean acts like a thermostat at $-2\,°C$, and the sea ice at $0\,°C$ while melting, the air temperature is most of the time below the freezing point. Consequently, the clouds can form solid phases almost all year long. Actually, Arctic cloud are in majority composed of ice, in a single solid phase or in mixed-phase (c.f. Fig. I.10). This almost systematic possibility of two phases existence makes their radiative properties hard to forecast. Consequently, the radiative impact of Arctic clouds can highly vary from a configuration to another, which makes the modelling of Arctic clouds and climate especially difficult.

Figure I.21 A shows the CRE of clouds at all latitudes. While in average clouds cool the surface, in the Arctic they rather warm it. The stability of the Arctic atmosphere helps them to have long lifetimes, which participate to maintain this warming effect. As for the MPCs, panel B indicates they have a cooling effect in summer, despite the fact in average, they warm the Arctic surface.

Opening Questions | Arctic climate

- What is the role of INP in the representation of Arctic clouds in models? (**??**)
- Is there an effect of INPs on the Arctic radiative budget through their interactions with clouds? (**??**)
- If so, what is its sign? (**??**)

./chap1/images/AA_feedbacks.png

Figure I.20: Illustrations of Arctic amplification feedbacks. Panel A is a schematics of the sea ice albedo feedback principle. First, a positive forcing induces an increase in temperature. This causes the melting of sea ice and snow, which leads to a decrease in albedo. Short wave radiation is then more strongly absorbed by the surface, which amplifies the temperature increase induced by the initial forcing, closing the positive feedback loop. Panel B: schematics of the lapse-rate feedback effect. The inversion of the vertical temperature profile induced, in the case of a surface warming of ΔT_{surf}, a negative buoyancy **B** below the inversion altitude. The heat gain is then poorly transported to the upper layers of the atmosphere, and the warming ΔT_{TOA} at the tropopause is minimal.

./chap1/images/matus_CRE.png

Figure I.21: Zonal mean of the cloud radiative effect on the surface for different types of clouds (panel A). Liquid clouds (red line), ice clouds (blue line), mixed-phase clouds (purple line) and multi-layers clouds (green line). Panel B shows the annual (black line), summer (red line) and winter (blue line) of the zonal mean CRE of mixed-phase clouds. Figure adapted from **matus_role_2017**

I.6 General intention and plan

The particles that compose clouds are hydrometeors which can be liquid or solid. The liquid cloud particles are water droplets, very opaque and reflective; the solid particles are ice crystals with complex geometry causing anisotropic scattering: they are more transparent. In the Arctic, there exists mixed-phase clouds that are composed of a combinaison of droplets and ice crystals, in an unstable equilibrium. Their radiative effect thus gets as complex as the microphysics that drives their formation and evolution.

Among dynamics and thermodynamics, a third type of actor is central in these processes: aerosols particles. They split in two families: CCN and INPs. The firsts allow the formation of droplets; the seconds impact the heterogeneous nucleation of ice crystals. Particles that can act as INPs are much scarcer than CCN, especially in remote regions, and therefore could have a larger role in determining the liquid-or-ice phase repartition in clouds. INPs can originate from a great spectrum of processes, but natural sources are known to predominate for cloud formation. In the Arctic, the knowledge on the nature and sources of INPs gets better as the research on it intensifies, but still has many gaps. Understanding the properties of INP populations, their abundance, seasonality, how they efficiently nucleate ice, is essential in order to accurately represent Arctic clouds in models. If modellers get the cloud phase repartition wrong, they inevitably misrepresent the energy balance of their modelled atmosphere. Consequently, since this deeply impacts the whole Arctic system, this puts in question all of our modelling studies of regional climate processes in the Arctic.

In this first chapter, we took a wide look at the state of the earth's climate. By highlighting the evidences of its global warming, and by exploring the underlying causes, we brought to light the importance of clouds. The description of cloud physics opened the topic of aerosol particles: these liquid or solid particles suspended in the air and that have the ability to craft clouds out of the blue, and to change their evolution pathways. We explored the specificities of the Arctic climate which will be the background region of the studies presented in this thesis. The fact it warms so fast is the expression of its high sensitivity to climate changes, and contributes to the difficulty to correctly represent it in climate models.

All the questions raised throughout the sections have led to the following central issue, around which this thesis revolves:

What is the causal relationship between local aerosol emissions, aerosol-cloud interactions and the rapidly warming Arctic climate, and do they constitute a feedback loop? and in particular **how do the aerosol particles populations responsible of the Arctic clouds formation will evolve in a warming climate?**

Although this thesis will not give a definitive answer to this question, it will lay the groundwork for answering it, following a broad investigation plan that tackles every link of this hypothetical feedback loop. First, we will show the development and evaluation of a new method combining observation and modelling tools, for the identification of the emissions sources of observed Arctic aerosols and especially INP. The method will be applied on various Arctic INP datasets in order to get clues on their likely emission sources (**??**).

Once the most important sources of Arctic INPs are identified, will be tackled the question of how to model these aerosol emission sources, and especially high latitude sources that will be shown to be crucial, and we will try to understand how they contribute to the observed populations of INPs (**??**). It will be the opportunity to evaluate different methodological approaches to represent ice nucleation, which will lead us to introduce an original way of representing–in a regional climate model–INPs from particle emissions (**??**).

In a third study, we will get to the actual impact of INPs on Arctic clouds. Their effect on the cloud phase, precipitation, lifetime and occurrence will eventually lead to the evaluation of INPs impacts on the Arctic radiative budget (**??**).

Eventually, Fig. I.22 is a schematic representation of this plan, that I endeavoured to follow all along the past three years, through a research work that I am happy to present in this thesis.

./chap1/images/mixed-phase-clouds.png

Figure I.22: Illustration of the general question: Studying the links and the potential feedback loop between INP emissions, aerosol-cloud interactions and the Arctic radiative budget, and what would be its sign?

II Conclusions and outlook

“Si tu peux, fais que ton âme arrive,
À force de rester studieuse et pensive,
Jusqu’à ce haut degré de stoïque fierté
Où naissant dans les bois, j’ai tout d’abord monté.
[...]
Fais énergiquement ta longue et lourde tâche
Dans la voie où le Sort à voulu t’appeler.”

Alfred de Vigny, *Les Destinées*

Chapter content

VI.1 Scientific context

Within Earth's climate, the poles are known to be extreme, yet very sensitive regions. The Arctic, with its central ocean, partly covered in sea ice and surrounded by lands of permafrost, is particularly vulnerable to the changes caused by global warming. Being located in the northern hemisphere, it shares this side of Earth with two thirds of the emerged lands, where nearly 90 % of the global population live. This exposes the Arctic to human activities, making it a region of a particular geopolitical importance. With a warming rate four times faster than the global average (**rantanen_arctic_2022**), the Arctic climate is both a complex and a very interesting system to study. Its rare conditions of solar exposition, atmospheric stability, temperature and remoteness (presented in Sect. I.5.1), make it especially difficult to observe and to represent in a numerical model.

The research project presented in this thesis aims to fill some of the knowledge gaps that strained our understanding of aerosols capable of forming cloud ice, as well as their impacts on Arctic clouds. The latter, which are mainly stratiform clouds at low altitudes, have in average a net warming effect at the surface, the opposite of mid- and low-latitude clouds (c.f. Fig. I.21). They are composed in majority of ice crystals or of a mix of ice and supercooled liquid droplets (c.f. Fig. I.10), which makes them especially sensitive to ice nucleating particles (INPs). Compared to cloud condensation nuclei (CCN)–which are sufficiently abundant to rarely be a limiting factor to cloud formation (**storelvmo_aerosol_2017**)–, INPs are scarcer by several orders of magnitude, which makes them determinant for cloud phase partitioning. A plume of INP can initiate the formation of ice crystals in a cloud, and helped by the Wegener-Bergeron-Findeisen (WBF) effect, can cause a radical change in the cloud phase. The associated change in the cloud radiative effect (CRE) can thus affect the whole atmospheric radiative budget (**storelvmo_aerosol_2008**). The misrepresentation of the effects of aerosols in atmospheric models is one of the reason for their poor depiction of clouds (**bony_clouds_2015**). The lack of knowledge on Arctic INPs–what are the species involved and what are their emission sources–, is one impediment to a better representation of INPs in models (**murray_opinion_2021**). Although the science of ice nucleation has been studied since the 18th century, accurately representing the formation of ice crystals in climate models remains a big challenge. Parametrisations of heterogeneous ice nucleation abound (**fletcher_active_1969**; **cooper_ice_1986**; **demott_predicting_2010**; **li_predicting_2022**), but they are often very specific to a region or to a given aerosol population, which makes difficult to generalize their use in climate models. Consequently, the models fail at keeping their role of efficient tools for the understanding of atmospheric mechanisms.

In this modelling thesis, I used the Weather Research and Forecasting model (WRF), its version that includes chemistry (WRF-Chem), and the FLEXible PARTicle dispersion model (FLEXPART) to explore these questions. My thesis research project consisted in three main steps: (1) getting a better understanding of the sources of Arctic INPs, (2) improving the modelling of particles that have ice nucleation ability and investigating how to accurately predict INP concentrations from them, (3) investigating the effect of INP on cloud properties.

VI.2 Sources of Arctic INPs

The first project of this thesis was the use of back-trajectory modelling to identify the geographical origin of INP observed in the Arctic. In order to achieve this, I started by assessing the performance of back-trajectory methods, and especially of the potential source contribution function (PSCF). The PSCF, or ratio method, is an analysis technique of Lagrangian particle dispersion model (LPDM) outputs that enables the identification of the likely emission sources of observed atmospheric species (**??**). It is more advanced than single back-trajectory analyses, which are extensively used in the atmospheric sciences community, but it has to our knowledge never been evaluated directly. To evaluate the PSCF method, we set up, in WRF-Chem, modelled emissions of idealised atmospheric tracers, and interpolated their concentrations as time series all around the Arctic in order to mimic in situ measurements. Then we used FLEXPART-WRF to compute Lagrangian back-trajectories, and finally applied the PSCF to these simulated measurements (**??**). The emission sources were already known, since they were modelled, which enabled the precise evaluation of the source identification method (**??**). However, the assessment results showed that the outputs of the PSCF method can be wrong and misleading, often indicating spurious origins for many species (**??**).

In order to construct a new, more precise method that could be used to identify the origin of Arctic short-lived species and especially INP, we conducted sensitivity tests on the parameters of the method. We ended up with the *improved ratio method*, that comprises the standard PSCF protocol with new parameters and additional treatments (**??**). Specifically, we found it was important (1) to correct the seasonal background in the concentration series, (2) to carefully define the percentile thresholds that identify high and low concentrations in the observation time series, (3) to define thresholds to exclude non-significant back-trajectories from the analysis, and (4) to combine the information of the likely origin of high concentrations given by the PSCF method with the negative of the likely origin of low concentration (where low concentrations in a species are unlikely to come from). Finally, we assessed the limitation of our improved method to better understand its results when applied to real cases.

In possession of a method with known performances, we were ready to undertake the identification of the sources of observed Arctic INPs. We used INP concentration data from three recent campaigns in the Central Arctic for different years and seasons (MOCCHA 2018, MOSAiC 2020 and ARTofMELT 2023, **??**), which comprised time series for different activation temperatures. The analyses revealed that INPs originate from inside the Arctic region, involving regional transport. Two main regions were highlighted: the coasts of the Barents and Laptev seas, and coasts of Greenland, with possible transport from inland Asia. These results suggest a predominance of mineral dust particles among the observed INPs, but the study of MOSAiC warm INPs (active at $-10\,°C$ and $-15\,°C$), showed possible oceanic sources from local biological activity near the coast, that themselves suggest the presence of marine organic particles in populations of warm INPs. The study of the origin of observed primary biological aerosol particles (PBAPs) during MOCCHA 2018 supported this conclusion, showing possible oceanic sources from ice free areas of the Arctic Ocean. These promising results motivate further use of the improved ratio method for the analysis of past and future INP datasets.

VI.3 Ice nucleation modelling and INP emissions

In **??**, we used this updated knowledge on the INP sources to improve the representation of aerosols particles capable of nucleating ice in WRF-Chem. Emissions of high latitude mineral dust (**??**), of marine particles organic aerosols (MPOA) from sea spray (**??**), and of continental PBAP from fungal spore and bacteria (**??**) were added to the model, satisfyingly reproducing the coarse-mode particles observations from MOCCHA 2018. The objective was to use the newly modelled particles to reproduce in situ INP concentrations in the central Arctic during the same campaign. This challenge has been addressed with two INP modelling approaches.

The first approach relied on the conversion, by ice nucleation parametrisations, of modelled aerosol particles to INP concentrations, after their transport in the atmosphere from their sources to the INP observation point. Eight different parametrisations were used; among them, three were specific to mineral dust particles, one to sea spray, two to bacteria, one to fungal spores, and one was general and used all types of coarse-mode particles (**??**). In addition to the modelled particles, observations from the MOCCHA 2018 campaign were also used to feed the parametrisations. The results showed that it is difficult to reproduce INP variability with modelled particles as input of the parametrisations (**??**). Nevertheless, concentration levels were correctly reproduced with the parametrisation that uses the whole coarse-mode (**demott_predicting_2010**). When the actual observed concentrations of particles were used to feed the ice nucleation parametrisations, the time variability was much better reproduced. The dust parametrisations applied to coarse mode observations overestimated the INP concentrations. This can be explained by the underlying assumption that observed coarse aerosols were composed of $100\,\%$ mineral dust particles. Therefore, our results, in agreement with other studies on the campaign, indicates that that the coarse-mode of the MOCCHA campaign is not made in majority from mineral dust (**??**). Concerning the biological particles observed during the campaign, the fluorescent PBAPs (fPBAPs), no INP parametrisation was able to activate enough of the observed PBAP to reproduce the observed INP concentrations. Indeed, the nucleation rate necessary for INPs to be driven by PBAPs is close to $100\,\%$ at $-20\,°C$. From the current range of our knowledge on ice nucleation with PBAPs, the latter can not explain alone the INP concentrations observed during the campaign. The fact fPBAPs are well correlated with the INP measurements suggests two things: (1) fPBAPs are transported along with other types of particles capable of ice nucleation, (2) fPBAPs observations underestimate the actual fPBAPs concentrations.

The second novel approach consisted in the direct emission and ageing of INP tracers from emissions of aerosols particles considered as potential INPs. With ice nucleation parametrisations suited to the particle species (dust, PBAP, MPOA), the aerosol emission flux was converted into an INP emission flux in five temperature bins, from $-5\,°\mathrm{C}$ to $-25\,°\mathrm{C}$. The INP tracers where then transported in the modelled atmosphere, subject to wet and dry deposition, as well as in-cloud removal by activation (**??**). This last process was the main mechanism that was missing in the first approach, in order to faithfully reproduce the INP nucleation history and the fact that efficient warm INPs are preferentially removed in cold clouds. These INP tracers were used to get insights on the spatial distribution of warm to colder INPs. Namely, it enabled the identification of zones where homogeneous nucleation can happen; paving the way for new implementation of INP-cloud interactions.

Interpolating the INP tracers concentrations at the location of the MOCCHA and MOSAiC campaigns made possible their comparison with INP observations (**??**). The variability range was better reproduced than what was obtained with the first approach, although the exact time variability of the observations could not be reproduced precisely. The average orders of magnitude were consistent with the observed INP concentrations and better than those of the first approach. These results provide confidence in this second approach of direct INP tracer modelling, and for this reason it was used further in the next chapter, to link INPs to the cloud microphysics for a better understanding of INP impacts on the properties of Arctic clouds.

VI.4 Adapting microphysics to better INP representation

The last step of my research project presented in this thesis was the investigation of the effects of INPs on Arctic clouds, using the WRF-Chem regional atmospheric model (**??**). I used the advances made in **??** on INP modelling to modify the model microphysics, using the predicted INP concentrations directly in the modelling of the immersion freezing of supercooled cloud droplets and the deposition mode nucleation of ice crystals. With two simulation configurations based on the Thompson aerosol-aware microphysics scheme (**thompson_study_2014**), the effects of using accurate Arctic INPs on cloud properties were quantified by the comparison of a control simulation (INPs produced by the **cooper_ice_1986** parametrisation, or C86) and a test simulation (with INP driven by the tracers of **??**).

The first comparison conducted concerned the concentrations of active INPs in the two set-ups. The comparison showed that the more physical approach of the new INP tracer simulation produces orders of magnitude less active INPs than the control simulation with C86. The C86 parametrisation tends to predict high concentrations over sea ice and Greenland because of low temperatures, even though these areas are far away from sources of particle potentially active as INPs. On the other hand, INP concentrations were much lower in the updated INP tracer run. Since the additional information on INP transport and removal was integrated, the spatial distribution was less homogeneous, and likely more realistic. Indeed, by assuming the validity of **??** about the INP tracers, and the correct diagnostic of temperatures by the model, the active INP concentrations of the INP-mp set-up should be a better representation of Arctic INPs than what provided the control simulation.

Then, the cloud properties derived from the two simulation set-ups were compared. In the new setup, using realistic INP tracers and lower INP concentrations, resulted in an unexpected large decrease of cloud cover. The difference is the strongest over the sea ice pack, where C86 is likely to overestimate INPs based on its temperature dependency The ice water path (IWP) showed a general increase, but a local decrease over Greenland and its northern coasts. Concerning liquid water path (LWP), a strong decrease was observed over Greenland and the whole sea ice region. Both effects need to be investigated in more details within a physical perspective, considering the strong decrease in active INP concentrations. We thus concluded that it is likely that these results could be due to limitations in the microphysical model, which is unable to predict ice cloud processes with such low INP concentrations.

The next steps will be to confirm this, and delve more deeply into these mechanisms in the microphysics scheme in order to improve them to set up a functional tool for the investigation of the Arctic cloud sensitivity to INPs. Simulating additional time periods for which observational data are available will make possible the evaluation of both the control and test simulations, providing valuable insights on the direction followed by our modelling approach. Finally, this study set-up will allow for a detailed investigation of the cloud phase partitioning and associated radiative impacts, crux of the thesis projects of other ongoing PhD work in my team at LATMOS.

The present thesis had to be concluded before this study could be brought to completion; nevertheless, it has opened the way for several promising avenues of research that are already being pursued.

VI.5 Perspective of future investigations

Whereas the work presented in this thesis has brought answers and insights to several research questions on INP-cloud interactions in the Arctic, it has also paved the way for many future investigations. The perspectives arising from this PhD work can be classified into three categories: continuations and extensions of the present study, avenues for further exploration, and openings toward more general questions.

Continuation

The improved ratio method for the identification of particle emission sources presented in **??** calls to be used with more INP datasets. In my analysis of INP sources, I did not make use of the entire MOSAiC 2020 campaign, focusing only on the cold season. Understanding the sources of the INP in spring and summer would be important to unveil the nature of warm-season INPs. It would also complement the results of the MOCCHA 2018 campaign analysis, and the comparison with the winter sources would give important insights on the seasonality of Arctic INP sources.

Beyond data from ice breakers campaigns, the development of more systematic long duration INP measurements with consistent instrumentation sets the conditions for better source detection. For example, the Portable Ice Nucleation Experiment (PINE) worldwide developing network is very promising (**mohler_portable_2021**). PINE is an automated and portable instrument that measures INP concentrations. Since a few years, it has been installed in many research stations around the world, including at high latitudes. The use of long duration measurements with backward modelling source retrieval methods would enable the collection of unprecedented information on INP sources and their seasonal variability. The application of the new ratio method developed in this thesis on a network of simultaneous measurements performed with the same instrument would be a world first in the study of INP sources.

On longer time scales, the Tara Polar Station project[1] was launched this year with the objective of ten missions of eighteen months each in the central Arctic Ocean. It will provide INP concentrations measured across the Arctic Ocean over twenty years, giving insights on their spatial and temporal evolutions within the scales of global warming. These expeditions will enable hitherto unseen data for the detection of INP sources.

Further exploration

The two side-studies presented in **??** showed the potential of our new ratio method in being adapted and applied in various conditions. The study on the origins of cloudy air masses encourages further investigations by raising questions: is it relevant to categorise clouds by their origins? Is cloud phase dependant on air mass origin? Are mixed-phase clouds originating from particular transport patterns?

In **??**, we set up updated simulated particles emissions with the objective of improving the INP modelling, but we did not use it in order to test the sensitivity of INPs to specific particle species. By switching on and off the different species that contribute to the INP tracers concentrations (**??**), it would be possible to quantify the contribution of each sources to the INP populations, getting information about their impact on the spatial repartition and on the temperature spectrum repartition. It motivates more simulation experiments, which could fit in a stand-alone study.

A misalignment of this thesis stood between the results of **??** and the set-up of **??**. In **??**, coasts of Greenland were shown to be intense sources of INPs; their topography and soil characteristics

[1]More information about the project can be found on the Tara foundation website: `https://fondationtaraocean.org/goelette/tara-polar-station/`.

suggesting mineral dust emissions. However, in **??**, the work on which we base the mineral dust emissions could not account for most of these Greenland sources. This is due to limitations in the land-use classification used in the WRF-Chem model which does not accurately represent the repartition of bare soil in this rapidly changing region. An important next step in INP modelling, would thus be the improved implementation of the Greenland sources among the model particles emissions. Similarly, emissions of marine organic aerosols and continental organics in this study used relatively simple parametrisations. In future work, more up-to-date representations could be used or new parametrisations could be created using the wealth of new observations in the Arctic, to better represent these local sources of warm INPs.

As mentioned several times already, the study of **??** is meant to be continued. The remaining questions about the implementation of realistic INP concentrations within an aerosol-aware microphysics scheme, as well as the knowledge gaps on the representation of the impact of INPs on clouds regional climate, will be addressed by the ongoing work started in my thesis.

Remaining general questions

Finally, there remains broader questions that this thesis has not addressed, but for which it has set the stage. In the first chapter (Sect. I.6), a pivot question was announced, around which the studies led during my PhD articulated. It was about the possible causal relationship between the INP emissions, their impact on clouds, and the evolution of the Arctic climate. My work showed that a link exists between aerosols emissions, INP populations in the Arctic, and cloud properties, with possible strong implications for clouds in the central Arctic. The advances on the understanding of Arctic INP sources gave insights on their sensitivity to climate change. With continued warming and snow and sea ice retreat, it is likely that all of the local sources of INP investigated in this thesis will be more intense in the future. Nevertheless, the sign of the INP impacts on regional climate remain unknown, as well as the future evolution of INP emissions. The study presented in **??** is still an ongoing work which necessitates further investigations in order to draw solid conclusions. I will continue this research along with my fellow LATMOS PhD student, Yaël Legars, who's thesis project consists in improving the representation of phase partition and radiative effect of Arctic mixed-phase clouds. Our results will also fuel the work of Lucas Giboni, fellow PhD student at Institut des Geosciences de l'Environnement (IGE) in Grenoble and co-supervised by LATMOS. His thesis focuses on understanding the radiative forcing of aerosol-cloud interactions within the framework of the rapidly changing Arctic climate. His work will contribute to a better understanding of the retroaction loop of aerosol particles and their effect on climate through their impact on clouds: do Arctic aerosol-cloud interaction warm or cool the regional climate, and what is the associated impact on aerosol emissions?

Other questions were raised in the presentation of my research work, even though they were not in the scope of the thesis. Among them, how do secondary ice production (SIP) and primary ice production (PIP) compare in their importance for cloud formation and evolution in the Arctic? In my thesis, we focused on PIP processes, but SIP are suspected to be of great importance in mixed-phase cloud physics (**sotiropoulou_impact_2020**; **zhao_global_2021**; **zhao_primary_2022**).

Naturally, the study of the Arctic climate feeds interrogations about the Antarctic. Although we know that INPs are much scarcer in the Austral hemisphere (**xu_characteristics_2023**), questions remain on their sources, transport, seasonality, effect on clouds, and evolution in a warming and thawing Antarctica. This thesis provided some of the foundations for investigating these research questions.

Questions & Answers | Modelling Arctic ice nucleating particles

Original question	Thesis answer
Which tools are required for the identification of aerosols particles emission sources?	Backward Lagrangian modelling in association with measurement time series (in situ, airborne or satellite) and analysed with the improved ratio methodology, enables the reliable identification of INP emission sources, which can successfully be applied to other types of atmospheric compounds (**??**).
What are the emission sources of Arctic INPs?	INPs observed in the central Arctic are likely to originate from inside the Arctic region, with possible transport from inland Asia. Two main regions have been identified: the Russian coasts of the Barents and Laptev seas, and the coasts of Greenland (**??**).
What are the important aerosol species for the ice nucleation of Arctic clouds?	The main INP sources suggest a predominance of dust particles in the Arctic, but the study of MOSAiC INPs active at higher temperature ($-10\,°C$ and $-15\,°C$), has shown possible oceanic sources that suggest the presence of marine organic particles in populations of warm INPs (**??**).
How do the different ice nucleation representations compare in a regional model?	Modelled organic particles used with adequate ice nucleation parametrisations give plausible INP concentrations. Measured concentrations of fPBAP converted into INPs with ice nucleation parametrisations give too low INP levels in order to support the hypothesis of PBAP-driven INPs (**??**).
Is it possible to use a general parametrisation for Arctic INPs?	It is, but is likely to give spurious results, especially with temperature-dependent parametrisations (**??**).
Is it possible to faithfully model INPs in the Arctic with our current knowledge, and how to do it?	Emission of INP tracers is an efficient approach to reproduce consistent levels of INP concentrations. Concentrations of $10^{-1}L^{-1}$ are rare in central Arctic for INP active at temperatures higher than $-20\,°C$ (**??**). Once integrated with the microphysics it predicts lower concentrations than general parametrisations, especially in the remotest regions of the Arctic (**??**).
Is the presence of INPs in the Arctic important for the formation of clouds and for the modification of their properties?	The modelled clouds can be very sensitive to how INPs are represented. The way microphysics schemes handle INP can induce dramatic impacts on cloud fraction and cloud phase (**??**).
Is there a relevant effect of INPs on the Arctic radiative budget through their interactions with clouds?	The conducted work was not able to provide a reliable answer to this question. However, the suggested impact of INPs on cloud induces an important change of the cloud radiative budget at the surface (**??**).

C. Ma thèse en 180 secondes

In this appendix is presented the text of my participation to the French contest *Ma thèse en 180 secondes*. Both the original in French and the translated version are given down below. The video recording with English subtitles is availbale on the Internet at the following adress: https://www.youtube.com/watch?v=pGzU3e5XTz0.

French (original)

Un rayon lumineux traverse l'espace, court entre les astres,
Lorsque la Terre sur son chemin l'intercepte.
De l'énergie que ce rayon convoie, l'atmosphère a l'embarras du choix.
Elle en reflète une partie et transmet le reste à la planète.

Sous ce flux de chaleur, et pour ne pas surchauffer,
La Terre à l'équilibre se met à rayonner.
Mais gaz et particules jouent au plafond de verre,
Et s'ils sont en excédant c'est réchauffement par effet de serre.

C'est ça, en émettant du CO_2 par exemple, on opacifie l'atmosphère, mais que dans un sens ! Les rayons du Soleil entrent comme avant, mais le rayonnement infrarouge que la Terre émet vers l'espace pour se refroidir, lui, est bloqué, et la chaleur confinée. Dans cet échange d'énergie les nuages, qui s'interposent, sont déterminants. Prenez par exemple ce baigneur, sur la plage, qui sortant de l'eau sent les rayons du Soleil réchauffer sa peau. Soudain ! un cumulus — un gros nuage — passant par là projette son ombre sur le nageur ; il a froid ! Au contraire, une fois la nuit tombée ce même nuage, parce qu'il piégera les infrarouges, aura un effet réchauffant.

Alors pour prédire le climat, on ne peut pas se passer de chercher à comprendre ces nuages. Et c'est tout l'objet de ma thèse : comment ils naissent, évoluent, précipitent ou s'évaporent. Pour finalement savoir s'ils réchauffent ou refroidissent. Le hic c'est que les nuages ne se forment pas spontanément dans l'atmosphère...

Vous connaissez toutes et tous le miroir embué en sortant de la douche ? Eh bien les nuages c'est un petit peu pareil ; comme la buée ils ont besoin d'un support sur lequel se former. Dans l'atmosphère ils naissent sur de petites particules en suspension dans l'air, qu'on appelle aérosols. On en distingue deux types : les aérosols qui permettent la formation des gouttelettes nuageuses, qui composent entre autres le nuage de notre baigneur, et ceux qui forment les cristaux de glace des cirrus par exemple, vous savez ces hauts nuages filandreux qui ne font pas d'ombre.

En Arctique — la région que j'étudie — le réchauffement est 4 fois plus rapide qu'ailleurs. On y rencontre des nuages dits en phase mixte particulièrement importants pour le climat. Composés à la fois de gouttelettes et de cristaux de glace, ils sont très dépendants du type d'aérosols dans l'air. Mais les modèles numériques de prévisions climatiques ne prennent pas en compte les aérosols, et donc représentent mal ces nuages.

Pour corriger ce problème, je traque les aérosols de l'Arctique afin de mieux comprendre leurs provenances et leur natures : seront-ils plus ou moins nombreux sur une planète qui se réchauffe ? Cela me permet ensuite de développer — sur ordinateur — de tout nouveaux modèles de nuages. Je les teste et les ajuste en les comparant à de vraies observations, jusqu'à ce qu'enfin je parvienne à simuler des nuages qui naissent, évoluent et disparaissent comme ceux qu'on voit dans le ciel.

L'urgence du réchauffement climatique nous pousse à le comprendre, pour autant nos recherches ne peuvent pas nous dispenser d'engager des changements sociétaux profonds. Car ce n'est pas la planète qui est en danger, c'est nous ! Alors, donnons-nous la chance de nous sauver nous-mêmes.

English (translated)

A beam of light travels through space, racing among the stars,
Until the Earth intercepts it on its path.
The energy it conveys, the atmosphere can decide:
Reflecting a part of it and transmitting the rest to the planet.

Under this flow of heat, and in order not to overheat,
The Earth reaches equilibrium and begins to radiate.
But gases and particles play a glass ceiling game,
And if they are in excess,
It is warming by greenhouse effect.

There you are ! By emitting CO_2 we make the atmosphere opaque, but only in one direction! The Sun's rays enter as before, but the infrared radiation that the Earth emits to cool itself is blocked, and the heat is trapped. In this energy exchange, clouds that interfere play a crucial role. For example, take a swimmer on the beach, as he emerges from the water he feels the Sun's rays warming his skin. Suddenly, a cumulus—a big cloud—passes by and casts its shadow on the swimmer; he gets cold! Conversely, once night falls, this same cloud because it will trap the infrared radiation, will have a warming effect.

So to predict the climate, we cannot do without understanding these clouds. And that is the focus of my thesis: how they form, evolve, precipitate, or evaporate. To ultimately determine if they warm or cool. The catch is that clouds do not form spontaneously in the atmosphere...

You all know the foggy mirror when you get out of the shower? Well, clouds are a bit like that; like fog, they need a surface on which to form. In the atmosphere, they form on small particles suspended in the air, called aerosols. They are of two types: aerosols that allow the formation of cloud droplets, which make up our bather's cloud, and those that form ice crystals in cirrus for example, you know, those high, filamentous clouds that don't cast shadows.

In the Arctic – the region I study – warming is four times faster than elsewhere. Mixed-phase clouds, particularly important for the climate, are found there. Composed of both droplets and ice crystals, they are highly dependent on the type of aerosols in the air. However, numerical climate prediction models do not account for aerosols and therefore poorly represent these clouds.

To tackle this problem, I track Arctic aerosols to better understand their origins and nature. Will they be more or less numerous on a planet that is warming? Next, this allows me to develop, on computer, new cloud models. I test and adjust them by comparing them to real observations until I finally manage to simulate clouds that form, evolve, and disappear like those we see in the sky.

The urgency of climate change drives us to understand it, but our researches cannot exempt us from making profound societal changes. It is not the planet that is in danger, it is us! So let's give us a chance to save ourselves.

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“Transitur et augebitur scientia”

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RÉSUMÉ

Les noyaux glaçogènes (INP, pour *Ice Nucleating Particles*) sont des particules d'aérosols capables de favoriser la formation de cristaux de glace dans les nuages, etc.

MOTS CLÉS

Interactions aerosol-nuages – noyaux glaçogènes – nuages froids – Arctique – régions polaires – microphysique – modélisation – climat